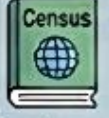
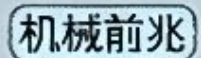
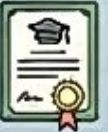


UPSC PRELIMS 2026: ELECTORAL RESET & FEDERALISM

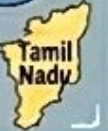
I 1. THE LOK SABHA EXPANSION (LEGAL PACKAGE 2026)

-  **Constitutional Basis:** Constitution (131st Amendment) Bill, 2026 amends Articles 11 &  
-  **The 2011 Census Pivot:** Delimitation decoupled from post-2026 census. Uses 2011 data.
- ★ **Seat Surge:** Increase strength from 543 to 850  →  
- ★ **Women's Reservation:** Precursor to operationalizing 33% quota by 2029.  

III 3. THE HYBRID MODEL: A WAY FORWARD

-  **Balanced formula** balancing “One Person, One Vote” with Federal Fairness. 
- ★ **Allocation Criteria:**  
- **50% Population:** Based on 2011/Current Census.
 - **50% Performance:** Based on State GDP (GSDP), Fiscal Contribution, Literacy/Health.
- ★ **Objective:** States with high exchequer/social indices rewarded political representation.  

II 2. STALIN'S CRITIQUE: “THE SWORD OVER SOUTH”

-  **Tamil Nadu CM M.K. Stalin & Southern leaders** raise Federal Alarm. 
- ★ **Punishment for Success:** Penalized for Population Control & high HDI. 
- ★ **Relative Power Dilution:** South share shrinks ~24% to ~19%. Voice silenced. 
- ★ **Demand:** Maintain 1971 relative weightage or use “pro-rata” increase. 

IV 4. KEY FEDERAL SAFEGUARDS (UPSC POINTERS)

-  Delimitation Bill, 2026 shifts “population” definition. 
-  “population as determined by Parliament by law”.
- ★ Grants significant political flexibility for redistributions.



ABHISHA AI by Abhishek Pathak

UPSC PRELIMS 2026: INDIA'S VULNERABILITY TRAP

I CONTEXT: WEALTH CONCENTRATION & THE SQUEEZE



- ★ **Macroeconomic Disconnect:** GDP growth (bar chart) vs Stagnant real well-being (flat line).
- ★ **Wealth concentration at Top** rising.
- ★ **Middle Class Vulnerability definition** (risk meter): Prob. of non-poor household falling below line due to risks (Health shock, Job loss, Inflation).



- ★ Limited creation of high-productivity formal jobs.

III ANALYTICAL FRAMEWORK: BEYOND INCOME (THE WB APPROACH)

A. THE WORLD BANK WELL-BEING SPECTRUM



- ★ Spectrum moves beyond binary metrics.
- ★ **India Challenge:** policy successful from 'Extreme' to 'Vulnerable', now need to move from 'Vulnerable' to 'Secure Middle Class'.

II CHALLENGE: POVERTY SHIFT & THE PRECARIETY TRAP



Current 2026



New Segment:

Households just above Extreme Line are ~30% (vs earlier 50%).

Shock

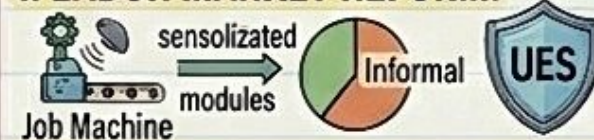


Extreme Poverty Rate: fallen (using \$2.15/day line).
Status: "Vulnerable," highly susceptible to ~30%.

- ★ **Precarity Trap**
Not in "extreme poverty," but no upward mobility.
- ★ Status: "Vulnerable," highly to falling back if crisis occurs.

IV THE WAY FORWARD: TO SECURITY & SHARED PROSPERITY

1. LABOR MARKET REFORM:

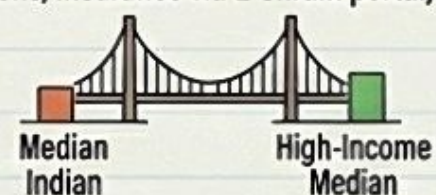


2. DE-RISKING HOUSEHOLDS:

Drastically reduce Out-of-Pocket Pensions, Insurance via E-Shram portal).

3. MSME ECOSYSTEM INTEGRATION:

Build Micro, Small, Medium Enterprise capacity (MSMEs create jobs). support with credit, technology, global value chains.



4. GLOBAL PROSPERITY GAP:

NITI MPM Score Falling (Refined focus on reducing precarity).

ABHISHIA AI
by Abhishek Pathak

UPSC CSE 2026: BALANCING DIGITAL ECONOMY & COSMOLOGY

DIGITAL LANDSCAPE (2026): REGULATORY BACKING & VALUE CLASSIFICATION

PART 1: DIGITAL ASSETS & BLOCKCHAIN (S&T)

1. DEPOSIT TOKENS



Concept: "Tokenized" form of bank account money. Unlike private crypto, these are regulated liabilities of a bank.
Backing: Traditional fiat (₹) in a regulated bank.
Example: Banks (e.g., JPM Coin) for instant client transfers.

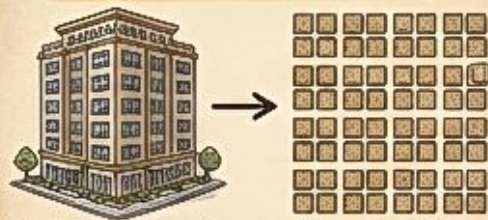
2. STABLE TOKENS (STABLECOINS)



Cryptocurrencies designed to have a stable price. Used as medium of exchange to avoid volatility.
Backing: Usually 1:1 by fiat (USD/INR), gold, or algorithm.
Example: USDT (Tether) = \$1 USD.

3. ASSET TOKENS

(RWA - TOKENIZED REAL-WORLD ASSETS)



Real Estate/Bonds → Property Tokens
Concept: "Fractional Ownership" (e.g., 1% of a building). **Backing:** Real-world assets.
Example: Rent share from fractional apartment ownership.

4. NON-FUNGIBLE TOKENS (NFTs)



Unique digital ownership certificates. Cannot be exchanged 1:1. Prove "Digital Scarcity".
Backing: Digital provenance (ownership history) on the blockchain.
Example: Digital Art (e.g., Beeple) or Metaverse Land.

PART 2: COSMOLOGY & THE EXPANDING UNIVERSE (GEOGRAPHY/PHYSICS)

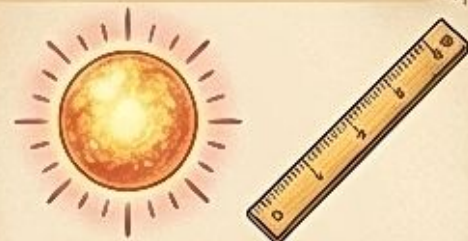
"CRISIS IN COSMOLOGY": HUBBLE TENSION & NEW PHYSICS

1. MEGAPARSEC (Mpc)



Unit of distance in extragalactic astronomy.
 $1 \text{ Megaparsec} = 10^6 \text{ parsecs} \approx 3.26 \text{ million light-years}$.
Importance: Denominator for universe expansion rate.

2. CEPHEID VARIABLE STARS



"Cosmic Yardsticks" for shorter distances.
Concept: Pulse rate (period) is directly related to true brightness.
Use Case: Find distance by comparing True vs. Apparent brightness.

3. TYPE 1A SUPERNOVA

White Dwarf



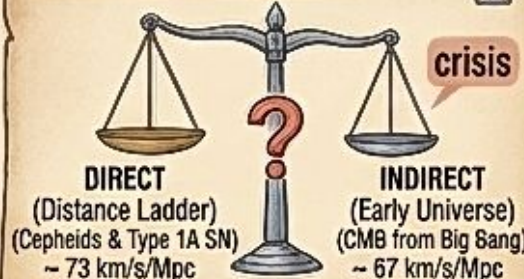
Chandrasekhar Limit $1.44 M_{\odot}$



Standard Candles

"Standard Candles" for long distances.
Concept: Occurs when a White Dwarf star explodes at a specific mass limit.
Use Case: Uniform brightness acts as a distance beacon.

4. HUBBLE TENSION



DIRECT
(Distance Ladder)
(Cepheids & Type 1A SN)
~ 73 km/s/Mpc

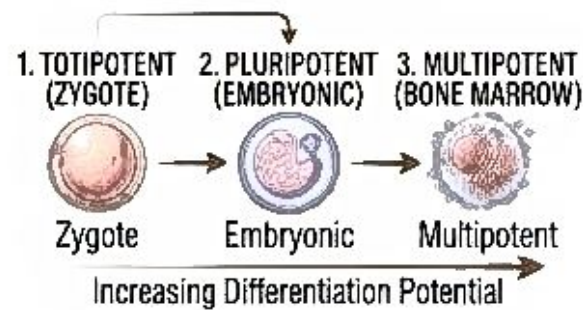
INDIRECT
(Early Universe)
(CMB from Big Bang)
~ 67 km/s/Mpc

Major Discrepancy in Hubble Constant (H_0).
Crisis: The 6 km/s/Mpc gap suggests missing Standard Model physics (e.g., Dark Energy, exotic particles).

UPSC CSE 2026: THE COMPREHENSIVE ECO-BIO CONCEPT REVISER

SECTION 1: STEM CELL TECHNOLOGIES & THE BIO-INNOVATION (S&T)

Progression 1: Understanding Stem Cell Potency



The Concept (Potency): A hierarchy of cell states, from maximum potential (Totipotent) to specific roles.

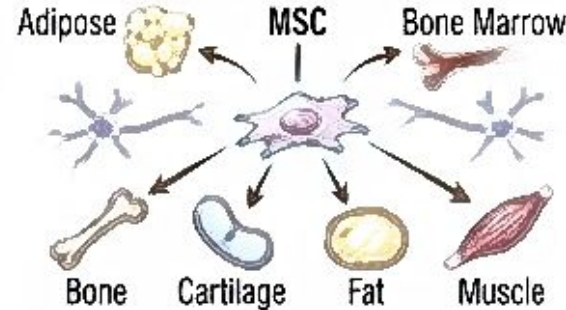
Signaling Mechanism: "Cellular Paracrine Signaling"—cell releasing small protein "whispers" to neighbors.

Factual Points: Focus on "Chandrasekhar Limit for Potency" metaphor, FDA-approved Leukemia treatments, regenerative medicine.

"Mesenchymal" vs. "Other" Stem Cells (Table)

Potency	Signaling	Harvesting
Multipotent vs. Pluripotent	High Immunomodulation	Non-Invasive
---	vs. variable	Non-Invasive vs. Invasive (Embryonic)

Progression 2: The Role of MSCs (Mesenchymal Stem Cells)

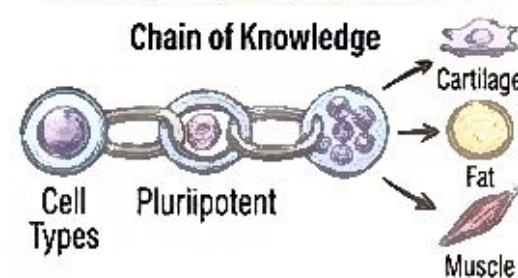


The Concept (Specificity): Key to success. MSCs from Bone Marrow & Adipose differentiate into specific tissues.

Signaling Mechanism: "Immunomodulatory Signaling"—MSC releasing "shield" cytokines to pacify an inflammatory immune cell (transplant rejection prevention).

Factual Points: "National Guidelines 2026" (regulatory context); sources: bone marrow, fat.

KEY CONCEPTS OF PART 1



SECTION 2: ECOLOGICAL SPECIES' ROLES & CONSERVATION

Progression 3: Keystone Species & Ecosystem Foundation

Keystone (e.g., Sea Otter)



The Concept (Foundation): Classical arch, Sea Otter is literal keystone. Removing it collapses Kelp Forest ecosystem. Shows "Low Biomass, High Impact".

Signaling Mechanism: "Trophic Cascade Signaling"—arrow system: Less Otter → More Urchins → Less Kelp. Population control.

Factual Points: Examples: "Apex Predators" and "Ecosystem Engineers" (e.g., Elephants creating water holes).

"Keystone" vs. "Sentinel" (Table)

Primary Value	Signal Type
Structural Stability vs. Population Control	Early Warning
	Early Stress Detection

Progression 4: Sentinel Species & Sensory Alerts

Alert (e.g., Canary in a Coal Mine)

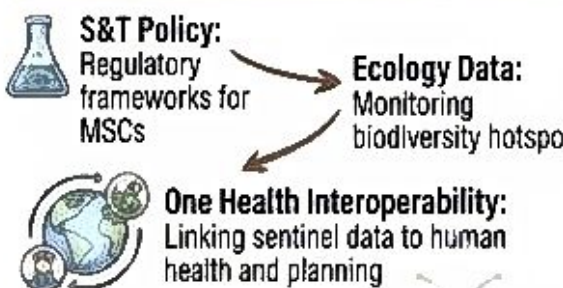


The Concept (Alert): Canary in mine provides "alerts" of danger (CO gas).

Signaling Mechanism: "Sensory Signaling"—a sensitivity-time graph showing Sentinel reacts sooner to toxin.

Factual Points: Examples for Air Quality, Frogs for Water Pollution, Vultures.

SECTION 3: STRATEGIC ACTIONS & INTEROPERABILITY ROADMAP (One Health)



STRATEGIC ROADMAP FOR NATIONAL COOKSTOVE INITIATIVE & CPCB STANDARDS

Part 1: NATIONAL STRATEGY FOR SUSTAINABLE BIOMASS (GS-3)

1. BIOMASS STOVER: RESOURCE & CHALLENGES



***Note:** "Stover" is defined as stalks and leaves.

2. MODERN FORCED DRAFT IMPROVED COOKSTOVE (GS-3)

Mechanism: Optimized gasification for total fuel conversion.

Outcome: Significant reduction in harmful indoor air pollutants.

2. MODERN FORCED DRAFT IMPROVED COOKSTOVE (GS-3)

1. Energy Potential: Essential for Next-Gen Biofuels.

2. Challenges: Open burning causes IAP & Black Carbon emissions.



BIOMASS COOKSTOVE COST ANALYSIS (FORCED DRAFT)

Cost Factor	Traditional Chulha	Modern ICS (e.g., Forced Draft)
Initial Capital	₹0 (virtually free)	₹2,000-₹4,000 (High initial Cost)
Operating Cost	₹0	Variable (e.g., pellets/processing)
Subsidy Dependence	Low	High (National Cookstove Initiative)



****INTEGRATED LIFE SCHEME FOR RURAL ADOPTION**

- Combining health, finance, and skill programs.

Part 2: ECOLOGICAL GUIDELINES FOR RITUAL OFFERINGS AND IMMERSIONS

"Crisis in Rituals": Environmental concerns & framework



Offerings: Flowers, idols, food, milk, oil.

The Concept: Rituals often have large ecological footprints.

Impacts: POP/lead paints → heavy metals; organic offerings → eutrophication.

Ritual Offerings vs. Sustainability Table

Offering Type	Env. Consequence	Mitigation/Sustainable Alt.
PoP Idols	Heavy Metal Poisoning	Seed Idols (grow into plants)
Organic Matter	Eutrophication	Artificial Folds for decomposition
Non-biodeg Items	Drainage clogging	Strict non-use/tines

***Note:** Reducing IAP is reserved for special cases where stalks and leaves cannot be treated.

4. REGULATORY FRAMEWORK & LEGAL PRINCIPLES

National Green Tribunal (NGT) Directives heavy fines for non-compliance.

CPCB Guidelines (2020): PaX Idol ban and artificial ponds requirement.



Judicial Precedents: Polluter Pays Principle (M.C. Mehta vs. UOI). (Update: Landmark UOI case established the principle).

Legal Basis: Water Act 1974, Article 51A(g) 51A(g) & 25 **Conflict:** Cultural Rights vs. Env. Duties.

INTEROP STANDARD DEVELOPMENT & POLICY INTEGRATION



Regulatory Alignment: NGT & CPCB Standard Harmonization

By Abhisha
AI by Abhishek Pathak
for UPSC CSE 2026

SPECIALIZED LEGAL TOPICS: HEAT WAVES & PIRACY LEGAL FRAMEWORKS

Part 3: HEAT WAVES & THERMAL INJUSTICE (GS-2/GS-3)

The 'Right to Cool' Framework

Constitutional Basis: Argument to recognize the Right to Cool under Article 21 (Right to Life and Dignity).



Structural Reforms:

- **Heat Action Plans (HAPs):** Moving beyond emergency response to urban design (cool roofs, urban forests).
- **Labor Laws:** Adjusting working hours for outdoor workers during peak heat.



The Legal Gap:

Currently, a Heatwave is NOT classified as a 'Natural Disaster' under the Disaster Management Act, 2005. This limits the release of central relief funds (SDRF/NDRF) compared to floods or cyclones.



Part 4: LEGAL CONSEQUENCES OF PIRACY (GS-2/GS-4)

Case Study: Leak of 'Jana Nayagana'

Issue: Digital piracy leads to massive revenue loss and infringes on moral/economic rights of creators.



Primary Legal Framework:

Copyright Act, 1957: Protects original literary, dramatic, musical, and artistic works.



Cinematograph Act, 1952: Recently amended to include strict penal provisions for unauthorized recording and exhibition.

Key Sanctions:

Criminal: Imprisonment (3 months to 3 years) and heavy fines (up to to 5% of production cost).



Civil: Injunctions (John Doe orders) to block websites/links before they even leak content.

Ethical Dimension:

Piracy as a violation of intellectual property and the 'sweat of the brow' doctrine.



By Abhisha
AI by Abhishek Pathak
for UPSC CSE 2026

UPSC PRELIMS 2026: ESSENTIAL TECH TERMINOLOGY

SECTION (1): MEMFLATION (Memory Inflation)

Term coined by Gartner: Rapid cost & demand inflation for memory chips (e.g., HBM) driven by AI.



DRIVERS: GenAI Boom, Large Language Models (LLMs).

CONSEQUENCE: Global shortage of High Bandwidth Memory.



IMPACT:

AI Infrastructure Bottleneck, Data Center Cost surge.

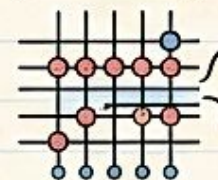


INDIA'S SEMICONDUCTOR MISSION

RAHUL BHARDWAJ
DNA SESSION: 16-04-2026

SECTION (2): MEMRISTOR & NEUROMORPHIC COMPUTING

► The 4th Fundamental Circuit Element (Resistor, Capacitor, Inductor... **MEMRISTOR**).



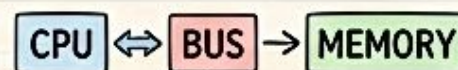
Crossbar Array: Hafnium Oxide (HfO_2) layer switching resistance.

Remembers state even with power **OFF**.

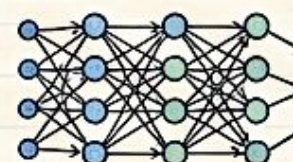
VON NEUMANN

vs.

NEUROMORPHIC



Energy & Time Waste



PROCESS & STORE IN-PLACE
SYNAPTIC EMULATION

► **BRAIN-INSPIRED SOLUTION:**
Mimics human synapses, Massive Parallel Processing, Ultra-low Power.

► **HISTORY:** Leon Chua (Hypothesized 1971)
→ HP Labs (Realized 2008).

KEY ADVANTAGES:
High Density,
Scalability,
Faster Processing.

ABHISHA AI
BY ABHISHEK PATHAK

(1) CARBON CAPTURE AND STORAGE (CCS): MINERAL CARBONATION

PERMANENT SEQUESTRATION via SOLID MINERALS

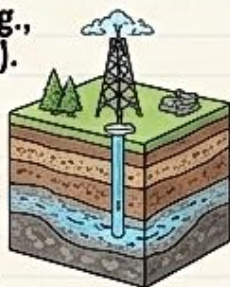
- The Concept: Reacting CO_2 with metal oxides (Ca, Mg) to form stable carbonate minerals like Calcite (CaCO_3) or Magnesite (MgCO_3).



- Comparison: **IN-SITU vs EX-SITU CARBONATION**

IN-SITU

- Location: Underground injection.
- Process: CO_2 reacts with natural reactive rocks (e.g., Basalt, Peridotite).
- Advantages: Large scale potential, direct storage.
- Example: CarbFix (Iceland).



EX-SITU

- Location: Above ground processing.
- Process: Reacting CO_2 with mined minerals or industrial wastes.
- Feedstock: Serpentine, Asbestos tailings, Steel slag.
- Advantages: Controlled environment, utilizes waste.
- Challenges: Higher cost, materials handling.



- SOLID MATERIALS for Carbonation:**



Natural Minerals
(Olivine, Serpentine)



Industrial By-products
(Fly Ash, Steel Slag)



Key Advantages of Mineral Carbonation:

- Permanent (Zero Leakage Risk)
- Environmentally Stable
- Circular Economy (Wastes to Resources)

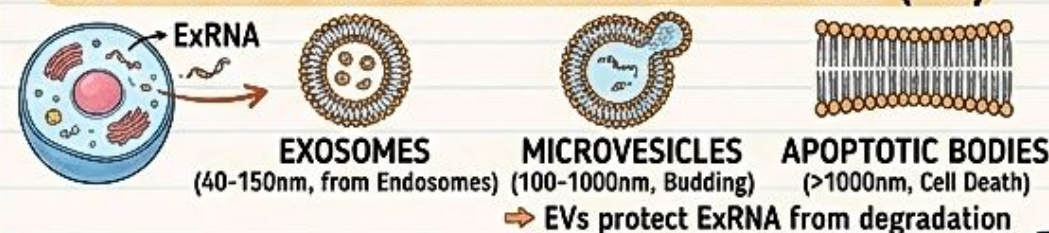
UPSC PRELIMS AND USES OF INDG RATH

(2) ExRNA ON A MISSION: EXTRA CELLULAR VESICLES



CELL-TO-CELL COMMUNICATION & APPLICATIONS

- What is ExRNA?
Extracellular RNA: RNAs found outside the cell, stable in body fluids.
- THE DELIVERY VANS: EXTRA CELLULAR VESICLES (EVs)



- TISSUE REPAIR AND USES OF ExRNA

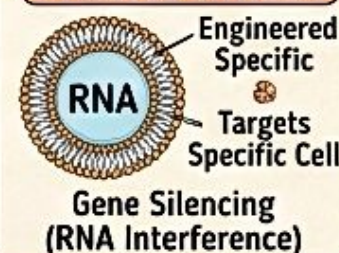


- Key Clinical Applications:

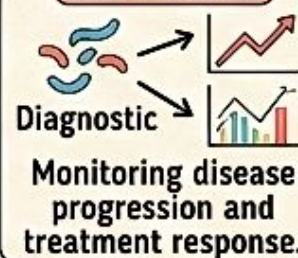
LIQUID BIOPSY



TARGETED THERAPY



BIOMARKERS



UPSC PRELIMS IMPOTANT POINTS:

- ExRNA Communication Consortium (ERCC)
- Significance in Personalized Medicine

UPSC PRELIMS 2026: KEY TOPICS - REVISION NOTES

Rahul bhardwaj DNA session 16-04-2026

1. DF-PIBM (Dynamic Floral-Pollen Interaction Biomechanical Model): WIND-INDUCED POLLEN

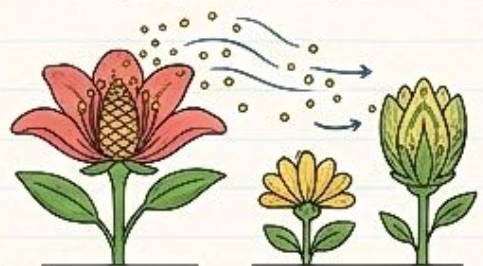


FINDINGS

- Aerodynamic Filtering:** Floral architectures (e.g., shape of pine cones, catkins) create turbulence to optimize capture.
- Vibration Resonance:** Wind causes male organs to vibrate at specific frequencies for timed pollen release.
- Settling Velocity:** Tuned grain size & density for local fluid dynamics.

FINDINGS

- Aerodynamic Filtering:** Floral architectures (e.g., shape of pine cones, catkins) create turbulence to optimize capture.
- Vibration Resonance:** Wind causes male organs to vibrate at specific frequencies for timed pollen release.
- Settling Velocity:** Tuned grain size & density for local fluid dynamics.



SIGNIFICANCE

- Agriculture:** Impacts yield prediction of wind-pollinated crops (e.g., maize, wheat) with climate change.
- Biodiversity:** Understands pollen rain in fragmented forests.
- Public Health:** Aids in predicting peak 'pollen bursts' for allergy management.



2. IMD MONSOON PREDICTION (Below Normal & Drivers)



BELOW NORMAL CLASSIFICATION

When seasonal rainfall is 90% to 94% of the LONG PERIOD AVERAGE (LPA).

LONG PERIOD AVERAGE CRITERIA TABLE

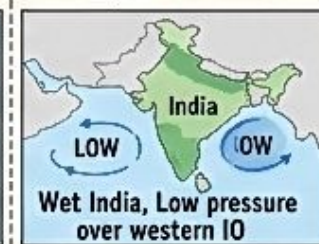
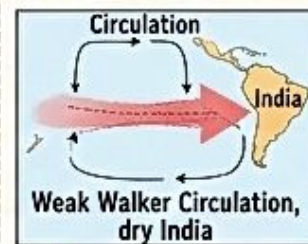
Category	Percentage of LPA
DEFICIENT	< 90%
BELOW NORMAL	90% - 94%
NORMAL	96% - 104%
ABOVE NORMAL	104% - 110%
EXCESS	> 110%

Current LPA (1971-2020 data) = 868.6 mm

EL NIÑO & IOD (The 'Tug of War')

EL NIÑO EFFECT: Abnormal warming of eastern Pacific, suppresses Indian Monsoon.

POSITIVE INDIAN OCEAN DIPOLE (IOD): Warmer western Indian Ocean, boosts Indian Monsoon, can counter El Niño.



Negative IOD exacerbates El Niño drought.

3. HUNGARY'S ELECTION & KLEPTOCRACY



KLEPTOCRACY DEFINED

Literally 'rule by thieves.' A government whose corrupt leaders use political power to appropriate nation's wealth, typically for embezzlement or misdirecting state funds.

KEY ASPECTS

- CRONY CAPITALISM:** Strategic sectors (media, construction) handed to loyalists/oligarchs.
- EROSION OF CHECKS & BALANCES:** Capturing judiciary & anti-corruption bodies to ensure impunity.
- WEAPONIZATION OF LAW:** Using legal frameworks to harass opposition & media ('Lawfare').
- TRANSNATIONAL MONEY LAUNDERING:** Use of offshore accounts, shell companies for illicit gains.
- POPULIST RHETORIC:** Masking rent-seeking behind strong nationalist/sovereignist discourse.



1. DAY-NRLM (National Rural Livelihoods Mission) - REVOLUTIONIZING RURAL INDIA



KEY ACHIEVEMENTS (as of April 2026)

- Mobilized over 105 Crore hoads



MAJOR DRIVER OF WOMEN & RURAL TRANSFORMATION

- Mobilized over 10.05 Crore households
- Formation of ~91 Lakh Self-Help Groups (SHGs)
- Focus on Financial Inclusion, Social Capital, & Livelihoods
- LAKHPATI DIDI initiative: Target 3 Crore women earning ₹1 Lakh+ annually
- LAKHPATI DIDI initiative: Target 3 Crore women earning ₹1 Lakh+ annually
- Transitioning from 'Subsistence' to 'Entrepreneurship'



WORLD BANK INTEGRATION & IMPACT

- Strategic partner: Technical Assistance & Credit
- Evaluations (2025-26) show:
 - Financial Inclusion: 50,000+ Bank Sakhis, facilitated ₹12+ Lakh Crore credit
 - Economic Impact: +28% increase in savings, ~20% reduced reliance on high-interest loans
- GLOBAL EXPORT: Model being adopted by Global South nations (e.g., Ethiopia, Tanzania)

2. INDO-BHUTAN CUSTOMS MEET - OPERATION NUMKHOR



Location: Munnar/Kochi, Kerala (April 2026)

PURPOSE: Address massive cross-border vehicle smuggling racket



THE ISSUE: Illegal import of premium 2nd-hand & luxury cars from Bhutan with forged docs



CRACKDOWN:



- Nationwide: Over 15,000 vehicles sharing identical chassis/engine numbers across Northeast & Kerala
- CRACKDOWN:** Nationwide operation by Indian Customs (Preventive) & DRI

KEY FINDINGS (CAG 2026): Over 15,000 vehicles sharing identical chassis/engine numbers across Northeast & Kerala

- Autora and carong auto loars s 2ane/luxury Cars
- MODUS OPERANDI:** Smuggled to avoid customs duty, registration via tampering VAHAN/PARIVAHAN database and collusion

3. INDIA'S CAFE-3 NORMS: THE GREEN ROADMAP



IMPLEMENTATION: From April 1, 2027

- OBJECTIVE:** Reduce carbon footprint, decrease crude oil import dependency
- MECHANISM:** Targets *Average* CO2 Emissions of a manufacturer's entire fleet
- IMPLEMENTING AGENCY:** Bureau of Energy Efficiency (BEE) under Ministry of Power (with Heavy Industries)
- GOVERNING ACT:** Energy Conservation Act, 2001
- APPLICABILITY:** M1 category vehicles (passenger <3.5 tonnes, <9 seats)

CAFE-2 vs CAFE-3 (PROPOSED)

FEATURE	CAFE-2 (CURRENT)	CAFE-3 (DRAFT 2026 / IMPLEMENT 2027)
Emission Target	~113 g/km CO ₂	78.9 g/km CO ₂ (Target by FY 2031-32)
Testing Cycle	Modified Indian Driving Cycle (MIDC)	Shift towards WLTP (World Harmonized)
Super Credits	1 EV = 3 cars	Retained (Incentivizes EVs & Hybrids)
New Features	Rigid targets	Carbon Credit Trading & Flexible Compliance

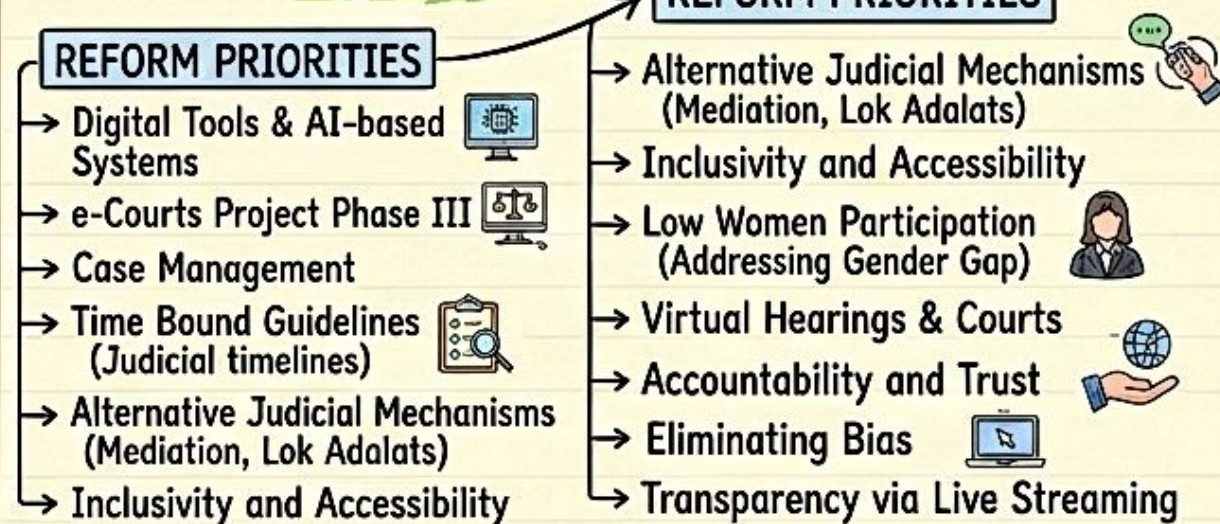
KEY CHANGES IN 2026 PROPOSAL

- Small Car Relief:** Eased curve for cars <1,229 kg for affordability
- Tightened SUV limits
- Weight Multiplier reduced: (0.002 to 0.00158) for initial compliance, but long-term path steep

UPSC TIP: CAFE = Fuel Efficiency (CO₂) | BS (Bharat Stage) = Pollutants (PM, NO_x, CO). Both critical for NET ZERO 2070.

CAUSES OF PENDENCY & DELAY

- Severe Backlogs (NJDG data: 55+ Million cases)
- Procedural Delay (Archaic laws)
- District Courts Pendency (Most cases here)
- Crisis of Pendency & Delay
- Long Trials
- Reduced Value of Outcome
- Erosion of Public Trust
- WHY DELAY
- REFORM



The diagram illustrates the technical causes of a boiler explosion. It features a central box labeled "TECHNICAL CAUSES" with five arrows pointing to a list of causes: Overpressure, Sudden Pressure Surge, Poor Water Management, Scaling & Tube Failure, and Negligent Operation. To the left, a detailed illustration of a boiler is shown with steam escaping from its top and front. A red arrow points from the boiler to a pink box labeled "CASE STUDY: BOILER EXPLOSION ON SHAKTI CHHATTISGARH". To the right, a small icon depicts a boiler with a fire inside and smoke coming out of the top.

CASE STUDY:
BOILER EXPLOSION ON
SHAKTI CHHATTISGARH

TECHNICAL CAUSES

- Overpressure
- Sudden Pressure Surge
- Poor Water Management
- Scaling & Tube Failure
- Negligent Operation

- Regulatory Gaps
- Inadequate Inspections
- Self-Certification Issues

- Self-Certification Issues
- Labour Concerns
- Contractual Labour (Unskilled, Unsafe)
- Skipping Maintenance
- Poor Training

- Critics argue the new Occupational Safety, Health and Working Conditions Code does not sufficiently hold top management accountable for contractor-level negligence.

The diagram illustrates the causal link between management's lack of accountability and a workplace accident. On the left, a hand icon points to a box labeled 'MGMT' (Management). To the right of 'MGMT' is a box labeled 'Insufficient Accountability'. A bracket connects 'Insufficient Accountability' to two boxes: 'CONTRACTOR' and 'Contractor-level Negligence'. A dashed arrow points from 'CONTRACTOR' to a yellow warning triangle icon with an exclamation mark, which is labeled 'accident' above it.

MODULE I: ECONOMIC GEOGRAPHY | THE CO-LOCATION ECOSYSTEM

Definition & Setup:

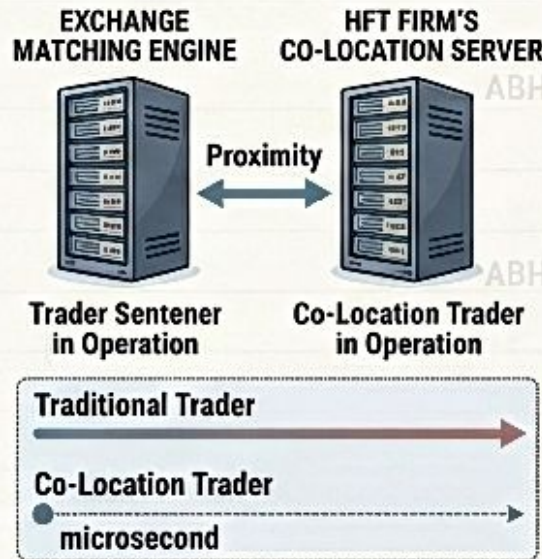
Exchange rents rack space inside **Exchange Data Centre** to members; houses servers in immediate **physical proximity hosting** to matching engine.

Microsecond Advantage:

Receive market data and execute orders in **fractions of seconds** (microseconds); essential for High-Frequency Trading (HFT) and Algorithmic strategies.

Market Dynamics:

- creates 'latency hierarchy'
- raises 'Fairness & Equal Access' concerns
- requires SEBI algorithmic trading guidelines
- potential for 'flash crashes'



MODULE II: MONETARY POLICY | MANAGING EXTERNAL SHOCKS

RBI Directive:

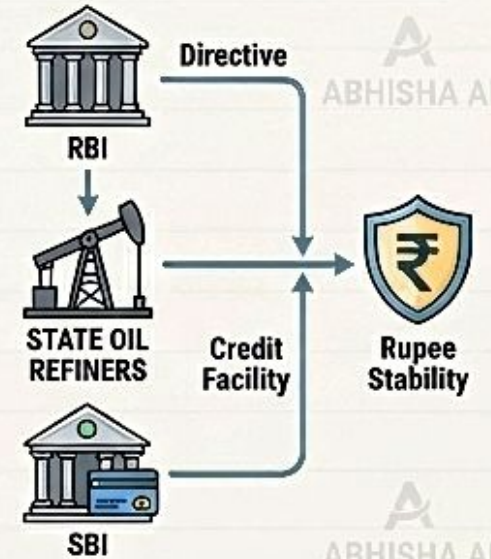
RBI asks state oil refiners to curb **spot dollar buying** during periods of Rupee volatility; reduces speculative pressure.

Alternative Forex Access:

Refiners are directed to a **SBI special exchange foreign credit facility**; provides USD without hitting open spot market; a type of capital control.

Objective & Stability:

Reduce immediate market pressure on INR; strategic move for **managing external shocks** (e.g., crude oil price spikes); crucial for maintaining exchange rate stability.



Module III: SCIENCE & TECH | THE GEN AI STARTUP PIPELINE

Institutional Context:

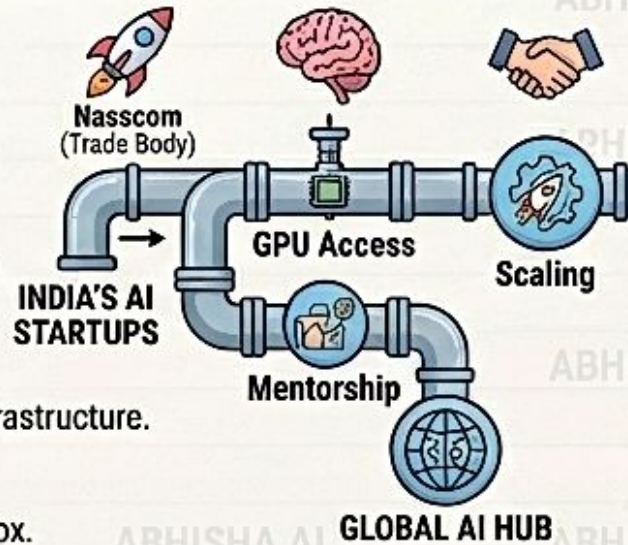
Nasscom (premier trade body for tech) launches **Gen AI Foundry**; fosters domestic innovation.

Strategic Goal:

Support **India's AI startups** build foundation models and applications; achieve sovereign AI capabilities.

Key Benefits:

1. Access to high-compute (GPU) infrastructure.
2. Technical & strategic mentorship.
3. Bridges research-to-market gaps.
4. Access to venture capital & sandbox.



Module IV: BIODIVERSITY | NATIONAL CHAMBAL SANCTUARY

Ecological Profile:

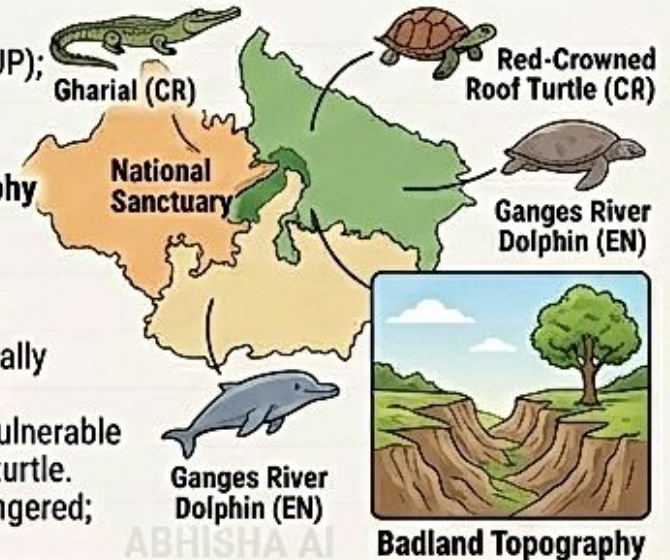
Unique tri-state sanctuary (RJ, MP, UP); vital riverine ecosystem.

Habitat Markers:

Defined by unique **badland topography** (ravines); part of **deciduous forest** eco-region; key nesting grounds.

Keystone Species (IUCN Status):

- **Gharial (CR)**: Stronghold for Critically Endangered fish-eating crocodile.
- **Red-Crowned Roof Turtle (CR)**: Vulnerable Critically Endangered freshwater turtle.
- **Ganges River Dolphin (EN)**: Endangered; national aquatic animal indicator.





ANALYSIS: POLITY & FAITH

 **Crown/Scale (Temporal Power) vs. Cross/Crescent (Spiritual Authority)**

- **Core Conflict:** Moral independence vs. State sovereignty.
- **Papal Checks (Historical):** Church's moral role to correct rulers.
- **Gallicanism (France):** King Louis XIV limiting Papal power, 'nationalizing' clergy.
- **Vessel Theology (US):** Political leader seen as divine tool; 'civil religion' risk.

WOMEN'S AGENCY PARADOX



Woman/Scale/Vote box

- **Statistics:** Voter Turnout (e.g., higher in 2024) vs. Low Representation (e.g., ~14-15% Parl, ~9% State).
- **Structural Barriers:** Patriarchy, financial constraints, male-dominated hierarchies.
- **Impact:** Local Bodies (PRIs) show female leaders prioritize social infrastructure (health, sanitation, education).

131st AMDT. BILL, 2026



Failed/2/3

- **Status:** Failed to meet 2/3 special majority.
- **Delimitation Deadlock:** Intrinsic link to future Census/seat redistribution.
- **Federal Concerns:** Southern states fear penalty for success; 'Trojan Horse' critique.
- **Way Forward:** Decouple reservation from delimitation; immediate implementation with JPC consensus.

ABHISHA AI

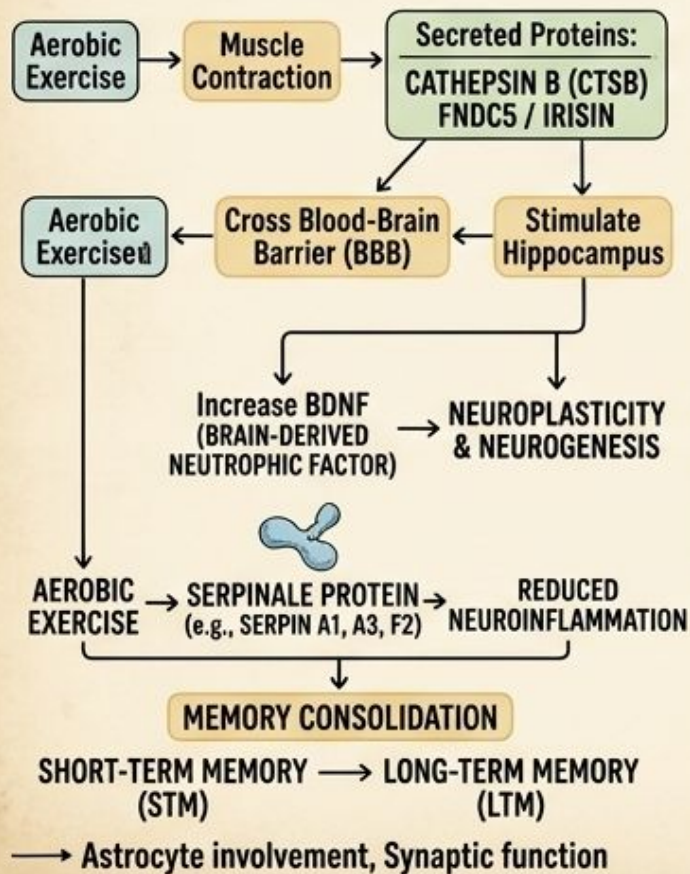
BY ABHISHEK PATHAK

RAHUL BHARDWAJ

DNA SESSION - 19/04/2026

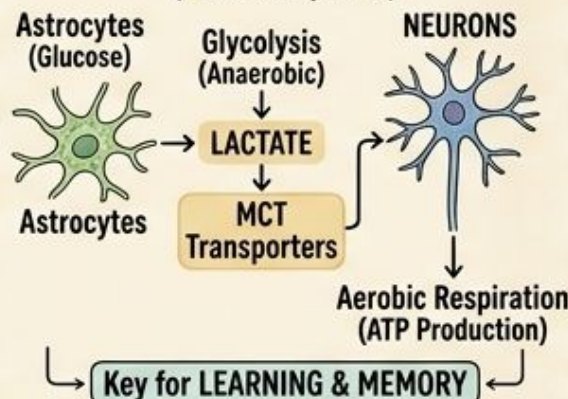
CURB 3 PROTEIN IN MEMORY LOSS & AEROBIC EXERCISE

THE MUSCLE-BRAIN CONNECTION (AEROBIC EXERCISE)

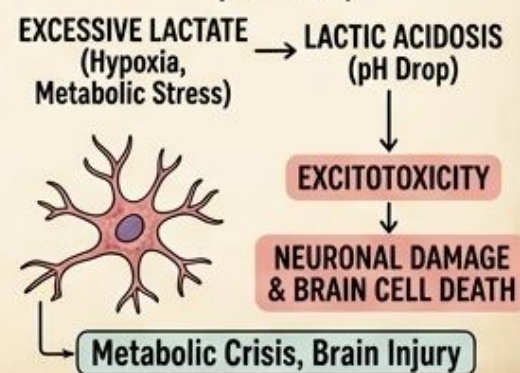


COGNITIVE FUNCTION & LACTATE

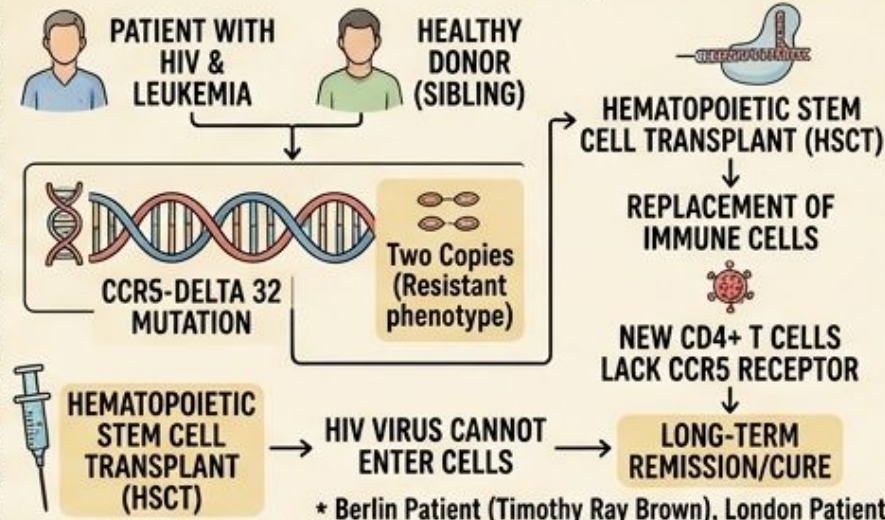
A. ASTROCYTE-NEURON LACTATE SHUTTLE (Positive/Optimal)



B. LACTATE NEUROTOXICITY (Excessive)



SIBLING TRANSPLANT & HIV CURE STRATEGIES



PSYCHEDELIC DRUGS OR HALLUCINOGENS

ALTERED PERCEPTION OF REALITY & COGNITIVE PROCESSES

DISRUPTING THE DEFAULT MODE NETWORK (DMN)

SUBSTANCE	ORIGIN/CATEGORY	MECHANISM OF ACTION	IMPACT
LSD (ACID)	Synthetic (Lysergic Acid)	SEROTONIN 5-HT _{2A} Receptor Agonist	Perceptual distortions, intense emotions
MAGIC MUSHROOMS	Natural (PSILOCYBIN)	Converted to PSILOCIN; Acts on 5-HT _{2A} Receptors	Alters perception, time sense, unity feelings
DMT	Natural (<i>Psychotria viridis</i>) also Ayahuasca	Binds 5-HT _{2A} & 5-HT _{2C} Receptors	Intense, short visual/auditory experiences
MESCALINE	Natural (Peyote Cactus)	Phenethylamine Alkaloid; Acts on 5-HT _{2A} Receptors	Traditional use, geometric visuals
AYAHUASCA	Natural Brew (DMT + MAO INHIBITOR)	Oral DMT activation via MAOI	Deep introspection, psychological healing

→ NEUROPLASTICITY

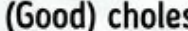
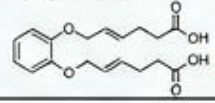
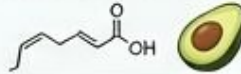
→ POTENTIAL TREATMENTS (Depression, PTSD)

→ CLASSIFICATION (NDPS Act)

UPSC PRELIMS 2026: COMPREHENSIVE STUDY NOTES

Rahul bhardwaj DNA session - 19/04/2026

1 GENERAL SCIENCE: BIOLOGY & NUTRITION

Feature	Saturated Fats	Unsaturated Fats	Artificial Trans Fats
Structure	• solid at room temperature	• liquid at room temperature	• hydrogen added to liquid vegetable oil to make it solid
Chain structure	• straight chain structure	• bent chain with double bonds	• Increases LDL  HDL
Structure	• increases LDL (Bad) cholesterol 	• reduces LDL • increases HDL (Good) cholesterol 	• Lowers HDL 
Examples	• butter, palm oil 	• olive oil, avocado 	• bakery goods Key Fact: Hydrogenation process

2 INTERNATIONAL RELATIONS: WEST ASIA DEMARCATATIONS



3 INTERNAL SECURITY: LEFT-WING EXTREMISM (LWE)

MHA Statistics - Peak (2010) vs 2025/2026



- Reason for Declines:
- Successful counter-insurgency ops like Kagar
- Developmental initiatives (e.g., roads, mobile towers) in tribal areas
- Winning hearts and minds through civic action

→ **Operation Kagar:** Massive multi-state offensive in Abujhmad, CG; records surrenders and eliminates leadership.

→ **Status/Concerns:** Government goal to eradicate LWE by March 31, 2026; risks of revival by shifting to MMC zone or Tri-junction.

4 SOCIAL ISSUES & LABOUR LAWS: NOIDA PROTESTS

Core Labour Issues: Violation of 8-hour workday, informal employment, lack of contracts, job security, social security.

Legal Framework: Minimum Wages Act 1948 (scheduled employments); Code on Wages 2019 (Floor Wage fixed by Centre, expands to all sectors).

→ **Governance Response:** Following 2026 Noida unrest, UP government announced 20-21% hike in minimum wages for Unskilled, Semi-skilled, Skilled categories.

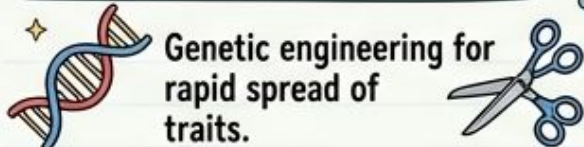


UPSC CSE PRELIMS 2026: CRUCIAL TOPICS OVERVIEW

ABHISHA AI
by Abhishek Pathak

Rahul bhardwaj
DNA session - 20/04/2026

(1. GENE DRIVE TECHNOLOGY & CRISPR-Cas9



Genetic engineering for rapid spread of traits.

- Uses CRISPR-Cas9 'molecular scissors'.
- Ensures >90% OFFSPRING inherit the gene.
- Enables specific genes to spread quickly in wild populations.
- APPLICATIONS:
 - Vector-borne Diseases: (e.g., modifying mosquitoes to prevent Malaria)
 - Invasive Species: (e.g., controlling agricultural pests)
 - Agriculture benefits (e.g., pesticide resistance)

3. BHARAT MARITIME INSURANCE POOL (BMIP)

- Domestic maritime insurance scheme.
- Backed by SOVEREIGN GUARANTEE.
- Primary for Indian flag vessels.
- Wide range of RISKS covered (e.g., war, environment).
- Initially run for 10 years.



2. ZERO-DAY VULNERABILITIES & AI IN CYBERSECURITY



Hidden security flaws in software/hardware.



- Unknown to the vendor.
- *Examples: Zoom flaws, Log4j.
- AI IN CYBERSECURITY:
 - Anthropic's Project Glasswing: Studying AI misuse (e.g., Claude mythos preview).
 - Uses of AI: Pattern recognition, threat detection.
- CONCERNS:
 - Risk of cartel-like control by a few large tech firms.
 - Centralized cartel structure in AI driven security ecosystem.
- *example project glasswing in AI driven cybersecurity.



centralized system

VS



decentralized ecosystem

3. BHARAT MARITIME INSURANCE POOL (BMIP)

- Domestic maritime insurance scheme.
- Backed by SOVEREIGN GUARANTEE.
- Primary for Indian flag vessels.
- Wide range of RISKS covered (e.g., war, environment).
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5. WEAKENING URBAN BREEZES

- Slower SEA BREEZE & LAND BREEZE.
- Impacts: Cool urban areas.
- Improves AIR QUALITY by flushing pollutants.



'normal'



'weakened' breeze

4. RBI PROPOSED KILL SWITCH



- For Card-based Payment Systems.

- Inspired by Singapore model.
- Enables disabling ALL DIGITAL TRANSACTIONS with one switch in case of fraud.

6. DIFFERENTIATING WELFARE AND DEVELOPMENT

WELFARE
(e.g., Subsidies, Cash transfers)



- Conceptual distinction.
- Immediate relief.
- Redistributive focus.

DEVELOPMENT
(e.g., Infrastructure, Education)



- Long-term structural process.
- Capacity building.

Trade-off

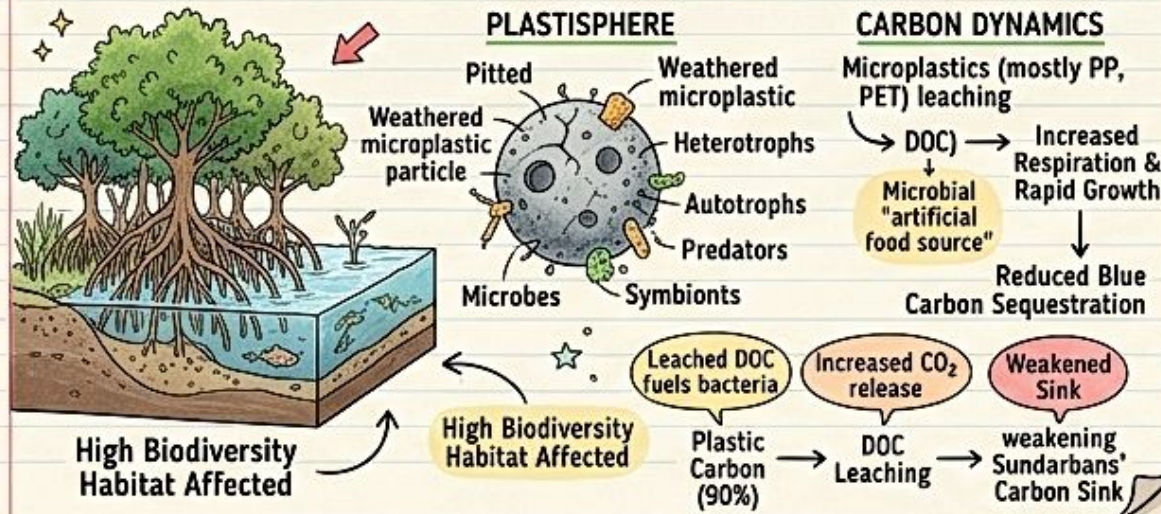


Balance

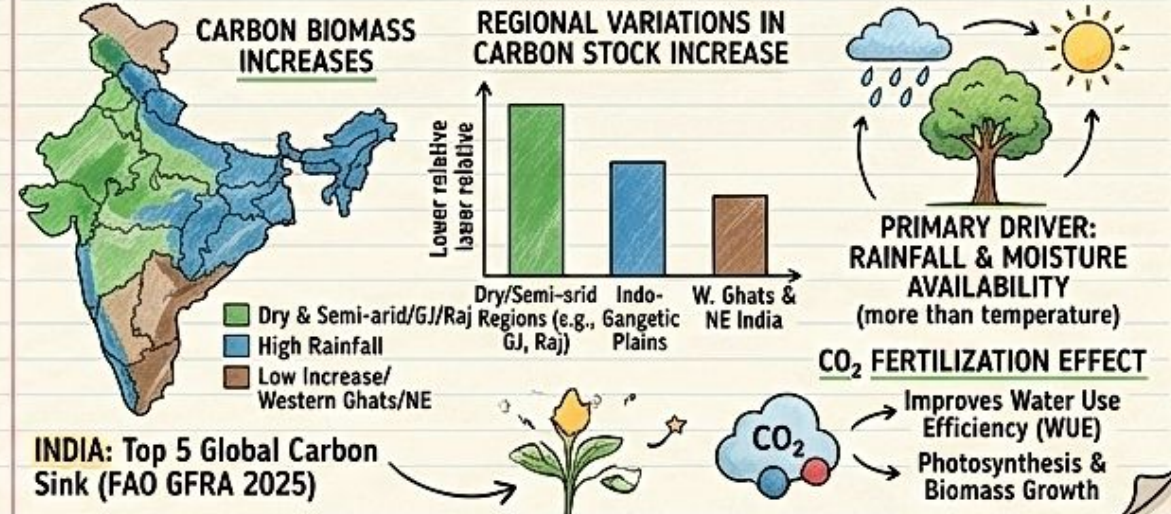
- Dangers of Populism/Partially Driven Transfers.
- Way Forward: Sustaining balanced growth.

UPSC CSE PRELIMS 2026: KEY TOPICS

1 SUNDARBANS: MICROPLASTICS ALTERING CARBON DYNAMICS



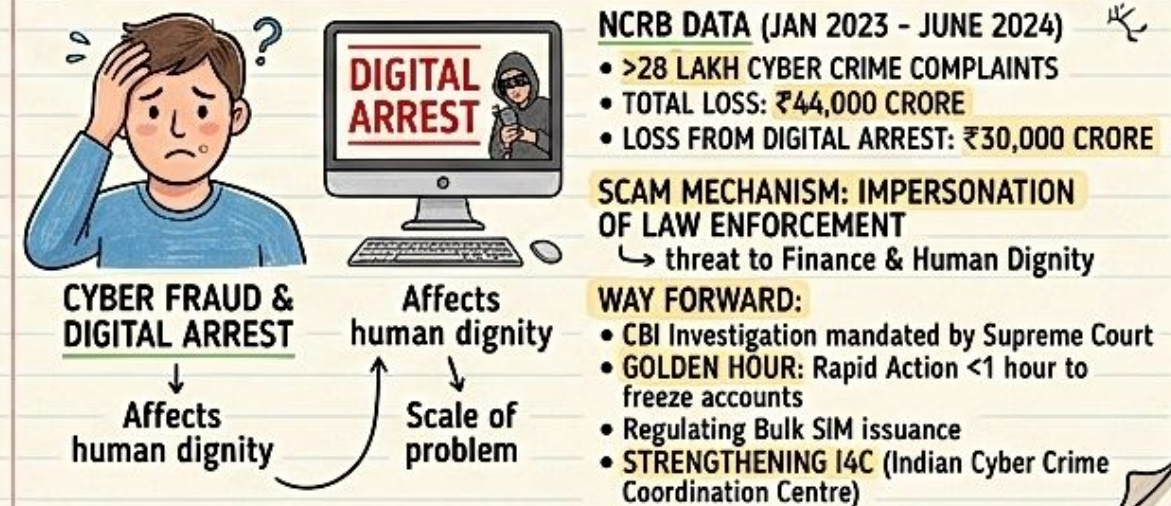
2 INDIA'S CARBON SINK: FOREST CARBON BIOMASS TRENDS



3 INDIA & KOREA CEPA: DEEPENING COOPERATION



4 CJI CONCERNS: CYBER CRIME & DIGITAL ARREST SCAM



1. SEBI & SOCIAL STOCK EXCHANGE (SSE) AMENDMENTS

SOCIAL IMPACT FUNDS (SIF):

- Cat I AIF
- Invests in NPOs/FPEs for measurable impact.

RETAIL PARTICIPATION



Minimum Investment Threshold (Individuals) → Slashed from ₹2 Lakh to ₹1,000

↓
Matches ZCZP minimum application.

ZCZP BONDS
(Zero Coupon Zero Principal)



NO INTEREST
(Zero Coupon)



NO PRINCIPAL RETURN
(Zero Principal)



Structured Donations.
Issued by NPOs on Social Stock Exchange (SSE)

KEY ACTORS

NPOs: Raise via ZCZPs, SIF donations, Development Bonds.
FPEs (For-Profit): Raise via Equity, Debt.

• 'INOPERATIVE' STATUS: AIFs with no remaining funds post-tenure can seek this status, reducing administrative compliance.

2. GLOBAL MARKETS: INVESTMENT NARRATIVES (2025-2026)

TINA: THERE IS NO ALTERNATIVE



Belief that US Equities (S&P 500) offer best returns.

Context: Strong US economy, energy shock resilience. Revived in 2026. (S&P 500 > 7,000)

TIARA: THERE IS A REAL ALTERNATIVE



Investors look beyond US to Europe & Emerging Markets (EMs) for valuations & diversification.

Shift: Weaker dollar made EMs attractive earlier. But recent outflows back to US suggest temporary eclipse of TIARA by TINA.

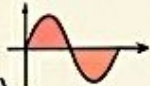
3. PHYSICS: INFRARED (IR) & QUANTUM SENSORS

Gamma Rays | X-Rays | Ultraviolet | **VISIBLE (Red)** | **INFRARED** | Microwaves | Radio Waves

Wavelength: ~700 nm to 1 mm

Nature of Infrared:

Transverse EM waves.
Invisible to naked eye, felt as Heat (Thermal Radiation).



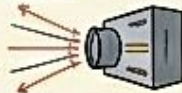
Quantum Interaction

- Wave-Particle Duality.
- Lower energy photons.
- Don't cause electronic transitions, but **DO** cause MOLECULAR VIBRATIONS (why absorbed as heat).



SENSOR SYSTEMS

Active Sensors:
Emit IR & measure reflection.
(e.g., LiDAR, TV remotes)



Passive Sensors:
Detect IR emitted by objects.
(e.g., Night-vision goggles, thermal imaging)



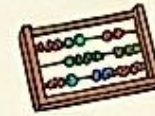
Energy

Quantum Sensors:
Utilize atomic quantum states to detect faint IR signatures. Crucial for **Deep-Space Astronomy**, **Stealth Detection**.

4. THE LEARNING CRISIS & FLN PUZZLE

LEARNING CRISIS:

Gap between 'Schooling' (Enrollment) & 'Learning' (Outcomes).



FLN: FOUNDATIONAL LITERACY & NUMERACY

→ Ability to read with meaning & perform basic math by end of Grade 3.

THE PUZZLE OF SALIENCE

Prominence/Urgency a problem holds.

- Child's inability to read is **INVISIBLE** compared to a broken road.
- Leads to low salience: → Not prioritized by Parents or Politicians



SYSTEMIC ISSUES

Low-Learning Equilibrium: Everyone satisfied with enrollment, despite absent quality.

Power Asymmetry: Poor parents lack 'voice'; Middle class has 'exited' to private schools.



WAY FORWARD

NIPUN BHARAT MISSION:
Universal FLN by 2026-27.

→ **MAKE LEARNING VISIBLE:**
Through Community-led assessments. Increases political & social salience.

UPSC CSE 2026: CRUCIAL GEOPOLITICAL & SCIENTIFIC UPDATES

1. PALEOANTHROPOLOGY: THE HAHNÖFERSAND FOSSIL CONTROVERSY



ELBE RIVER, GERMANY

- **ORIGINAL THEORY:** 75,000-year-old Neanderthal-Sapiens hybrid.
- **MODERN DISCOVERY:** Confirmed as Mesolithic *Homo sapiens*.
- **NEW AGE:** Approx. 7,500 years old.
- **SIGNIFICANCE:** Eliminates major "missing link" theory. Highlights phenotypic variation in modern humans.
- **UPSC LINK:** GS Paper I, Anthropology Optional.



3. INTERNATIONAL RELATIONS: DRUZHBA OIL PIPELINE NETWORK CRISIS

DRUZHBA OIL PIPELINE NETWORK



DRONE STRIKE NEAR BRODY HUB (UKRAINE)

- **WORLD'S LONGEST** crude oil pipeline, named 'FRIENDSHIP'.
- **STRATEGIC GEOGRAPHY:** Splits into Northern and Southern legs. Northern: to Poland, Germany. Southern: to Hungary, Slovakia, Czech Republic.
- **2026 CRISIS:** Conflict-related disruption in Ukraine halted Southern flow.
- **CONSEQUENCES:** Hungary's "Pipeline Diplomacy"; leveraged veto on EU aid to Ukraine for restoration.
- **UPSC LINK:** GS Paper II (International Relations - Energy Security).



2. ECONOMY: RISE OF SHAREHOLDER ACTIVISM IN JAPAN



ACTIVIST INVESTOR

- **CONTEXT:** Record surge in 2025-2026, shifting corporate culture.
- **CORE TACTICS:**
 - **PROXY BATTLES:** Challenging board elections and leadership.
 - **ASSET RATIONALIZATION:** Pushing for spin-offs of "non-core" assets.
- **GOALS:** Unlock value, increase returns, and improve governance.
- **DRIVERS:** End of deflation, Tokyo Stock Exchange initiatives (e.g., P/B ratio push).
- **UPSC LINK:** GS Paper III (Economy).



4. SECURITY: CANADA'S ARCTIC SOVEREIGNTY SHIFT



CANADIAN ARMY'S 'GO IT ALONE' AIM

- **CONTEXT:** Asserting greater capability, reducing reliance on US.
- **NORAD EVOLUTION:** Canada remains a partner in aerospace warning/control, but launched independent initiatives.
- **KEY INITIATIVES:** Operation NANOOK-NUNALIVUT (2026).
- **DEVELOPMENTS:** Arctic airport upgrades (e.g., Inuvik, Rankin Inlet) and improved Domain Awareness (e.g., in Northwest Passage).
- **INTENT:** Resilience against rising global interest in the Arctic (Russia, China).
- **UPSC LINK:** GS Paper II & III (Internal Security, IR).



UPSC CSE 2026: INTEGRATED GS STRATEGY - ENERGY SECURITY & INCLUSIVE GROWTH

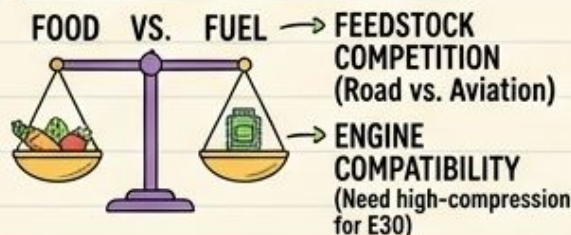
PART 1: BIOFUELS & INDIA'S ENERGY SECURITY IN 2026.



1. THE BIOFUEL LANDSCAPE: REDUCING 85% CRUDE DEPENDENCE.



SAF (SUSTAINABLE AVIATION FUEL):
1% SAF BLENDING BY 2027 (INTL. FLIGHTS)
PATHWAY: ATJ (ALCOHOL-TO-JET)



THE 2026 ENERGY CRISIS: A LESSON.

FEB 2026 WEST ASIA CONFLICT DISRUPTION



ENERGY VULNERABILITY

- Low LPG strategic reserves, supply slashed, price spikes

2. CHALLENGES IN BIOFUELS.



FOOD VS. FUEL (e.g., sugarcane diversion)
FEEDSTOCK COMPETITION (Road vs. Aviation)
ENGINE COMPATIBILITY (Need high-compression for E30)

3. CBG & THE RURAL ECONOMY (CBG: COMPRESSED BIO-GAS)

GOBARDHAN SCHEME



SATAT SCHEME



USES:

- CLEAN COOKING
- TRANSPORT (CNG Substitute)
- FOM (FERMENTED ORGANIC MANURE) FOR SOIL

WAY FORWARD:

- DIVERSIFY SOURCES
- SCALE CBG
- STRATEGIC LPG RESERVES

PART 2: INFORMAL WORKFORCE & INCLUSIVE GROWTH.



1. THE REALITY OF THE 'NEARLY 90%':

INDIA'S WORKFORCE (>90% INFORMAL)

TRIPLE BURDEN:

- INCOME INSECURITY
- LACK OF SOCIAL PROTECTION
- HIGH COST OF LIVING

3. SHIFT IN GOVERNANCE PARADIGM.

WASHINGTON CONSENSUS

- Deregulation
- Informalization

RIGHTS-BASED PROVISIONING

- Health, Education
- Social Security as entitlements

WHY HIGH INFORMAL EMPLOYMENT?

- LABOR MARKET DUALISM
- POOR EDUCATION/SKILLS
- BARRIERS TO FORMAL SECTOR ENTRY
- POPULATION PRESSURE

2. STRUCTURAL SHIFTS & URBAN/RURAL CRISES.



WASHINGTON CONSENSUS

- Deregulation
- Informalization

RIGHTS-BASED PROVISIONING

- Health
- Education
- Social Security as entitlements

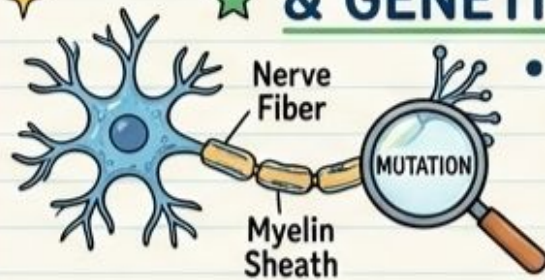
SOLUTION: THE KERALA MODEL



- DECENTRALIZATION (Panchayats)
- WELFARE BOARDS FOR SPECIFIC SECTORS
- HIGH HUMAN CAPITAL INVESTMENT (Health/Education as floor)
- UNIVERSAL SOCIAL SECURITY

*LINK ENERGY & LABOR IN ANSWERS, E.G., 2026 LPG CRISIS AFFECTED INFORMAL WORKERS

I. PERIPHERAL NEUROPATHY & GENETIC BASIS



- A diverse group of genetic disorders affecting nerves outside the brain and spinal cord (e.g., Charcot-Marie-Tooth - CMT).



★ KEY CONCEPTS:

GENETIC BASIS:

Mutations in genes for myelin (insulation) or axonal transport.

✗ NULL MUTATIONS (LOSS-of-FUNCTION): ✗

Genetic change causing a gene to be non-functional or not produced.



Examples:

1. Haploinsufficiency



(one good gene not enough)

2. PMP22 Gene



(one missing copy → HNPP)

★ OTHER KEY GENES: GJB1, MPZ, MFN2.

★ SUMMARY: Gene damage leads to nerve dysfunction.

II. JUDICIAL ACCOUNTABILITY & THE 'VARMA CASE'



Context: Justice Yashwant Varma (April 2026) resignation highlights accountability loop-hole.

1. LEGAL FRAMEWORK

- Articles 124(4) & 218: Removal for 'proved misbehaviour or incapacity'.
- Judges (Inquiry) Act, 1968: The Process.
100/50 MPs → Inquiry Committee → Special Majority Vote.

2. THE 'RESIGNATION' ISSUE

- Loop-hole: Judge resigns before parliamentary vote.
- Effect: Process becomes 'moot'. Judge avoids stigma & retains post-retirement benefits (e.g., pension).
- Impact: Erosion of Public Trust & Judicial Accountability.
↳ SC Published In-house Report (e.g., burnt currency allegations).

3. WAY FORWARD

- Revive Judicial Standards & Accountability Bill.
- Amend 1968 Act: Allow proceedings post-resignation to determine benefits.
- Establish permanent disciplinary mechanism.

1. INDIA-EGYPT DEFENCE COOPERATION (GS II - IR)

STRATEGIC PARTNERSHIP: BEYOND THE SUEZ

➤ 11th JOINT DEFENCE COMMITTEE (JDC) MEETING (CAIRO, APRIL 2026)

➤ BILATERAL DEFENCE COOPERATION PLAN (2026-27):

- ↳ Structured Military Interaction
- ↳ Joint Training & complex multi-domain Exercises

➤ DEFENCE INDUSTRY:

- ↳ CO-DEVELOPMENT & CO-PRODUCTION
- ↳ India pitched 'MAKE IN INDIA, MAKE FOR EGYPT' (Leveraging \$20B domestic capacity)



MARITIME SECURITY:

- ↳ NAVY-TO-NAVY STAFF TALKS
- ↳ INDIAN OCEAN REGION (IOR) & SUEZ CANAL (Strategic status as gateway)
- ↳ Egypt tracks threats via India's INFORMATION FUSION CENTRE-IOR (IFC-IOR)

3. ECONOMY CHALLENGES OF WAR (GS III - ECONOMY)

GLOBAL CONFLICTS & INDIA'S POLYCRISIS SCENARIO

➤ MACROECONOMIC IMPACT:

- ↳ CRUDE OIL: Imported Inflation. (+\$10/bar = +CAD ~0.5% of GDP)
- ↳ FUEL!: Imported Inflation(+\$10/bar = +CAD ~0.5% of GDP)
- ↳ FERTILIZERS: Disrupted NPK supply chains (Russia/Belarus). Spikes Food Subsidies, Food Security risks
- ↳ REMITTANCES: Instability in West Asia threatens inflows (~\$100B+ for India). Balance of Payments effect



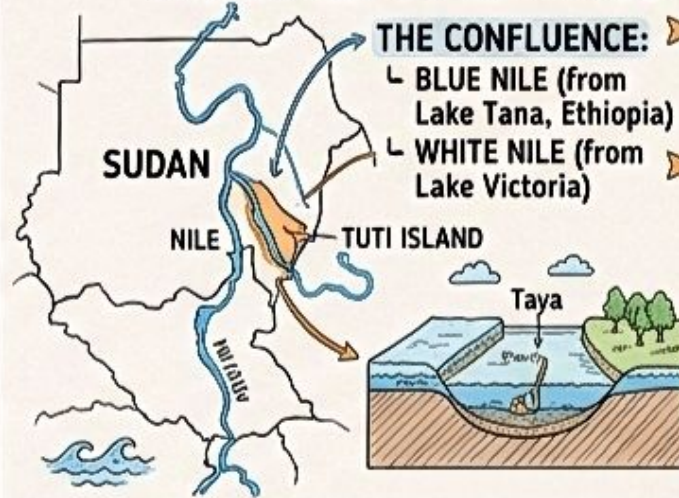
MICROECONOMIC EFFECTS:

- ↳ SUPPLY CHAIN SQUEEZE: Higher freight rates (e.g., Red Sea crisis). Increases 'Cost of Doing Business'
- ↳ GLOBAL DEMAND: Slowdown reduces EXPORTS for MSMEs (textiles, gems, jewelry)
- ↳ INPUT COSTS: Affects farmers via high diesel & fertilizer prices. 'PROFITABILITY CRISES' in agriculture



2. TUTI ISLAND: GEOGRAPHY OF KHARTOUM (GS I - GEOGRAPHY)

THE HEART OF KHARTOUM AT THE CONFLUENCE



THE CONFLUENCE:

- ↳ BLUE NILE (from Lake Tana, Ethiopia)
- ↳ WHITE NILE (from Lake Victoria)

FORMATION:

- ↳ Crescent-shaped RIVER ISLAND (~8 sq. km)
- ↳ Result of centuries of SILT DEPOSITION

'TAYA' SYSTEM (Indigenous early-warning):

- ↳ Residents use DRUMS & MONITORING TENTS to predict season surges
- ↳ Example of COMMUNITY-BASED DISASTER RISK MANAGEMENT (CBDRM)

STRATEGIC STATUS:

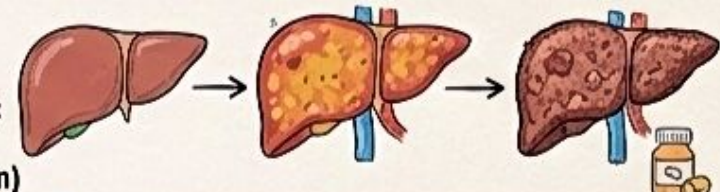
- ↳ 'RURAL EYE' in the city. Key for agriculture
- ↳ Threats: Climate change, upstream damming

4. LIVER FIBROSIS & NASH (GS III - S&T/HEALTH)

THE SILENT EPIDEMIC: FROM FATTY LIVER TO CIRRHOSIS

➤ PROGRESSION:

1. STEATOSIS (Simple fatty liver)
2. NASH (Non-Alcoholic Steatohepatitis: Inflammation + damage)
3. FIBROSIS (Excess scar tissue/collagen)
4. CIRRHOSIS (Advanced scarring, Liver Failure, HCC/Cancer)



➤ PREVALENCE RISKS:

- ↳ Strong correlation with TYPE-2 DIABETES (T2DM)
- ↳ BMI > 30 (Obesity)
- ↳ 25-30% global adult population has hepatic steatosis

THE 'LIFESTYLE' FACTOR & REVERSIBILITY:

- ↳ High fructose/ultra-processed food, low physical activity
- ↳ 10% Weight Loss = 90% NASH resolution & 45% fibrosis regression

➤ UPSC NOTE: Mention National Programme for Prevention and Control of NCDs (NPCDCS)

UPSC Tip: For 2026, link Tuti Island to 'Nile Water Conflict' (GERD) and Liver Fibrosis to 'National Health Mission' targets.

UPSC CSE 2026 FOCUS: GLOBAL HEALTH & CLIMATE ADAPTATION

PAGI 1: WHO PANDEMIC AGREEMENT

(78th World Health Assembly, 20 May 2025)



CONTEXT & AIMS

- rectify 'catastrophic failures' of COVID-19
- establish legally binding framework for 194 Member States

ONE HEALTH APPROACH



KEY MECHANISMS

'THE HEART OF THE AGREEMENT'

PABS

(PATHOGEN ACCESS & BENEFIT-SHARING)

- Guaranteed share of vaccines, e.g., 20%
- From shared samples

GSCL

(GLOBAL SUPPLY CHAIN & LOGISTICS)

- Network to prevent 'vaccine nationalism'
- Equitable distribution

COORDINATING FINANCIAL MECHANISM

- Pooled resources for preparedness in resource-limited settings

WAY FORWARD

- OPERATIONALIZING ANNEXES
 - Finalizing PABS technical details

CORE PRINCIPLES

- AIMS TO FIX GLOBAL FAILURE, EQUITY & COLLECTIVE SECURITY
- Integrated surveillance
- recognizing interdependence of human, animal, environmental health



CHALLENGES & INEQUITIES

- 摩擦 BINDING COMMITMENTS
 - Developed nations resist tech transfer, IP waivers
- THE 'HYBRID' DILEMMA
 - Immediate pathogen sharing vs deferred benefits
- SOVEREIGNTY VS. SURVEILLANCE
 - National rights vs international oversight

- RATIFICATION
 - Sustained political will to move from adoption to implementation

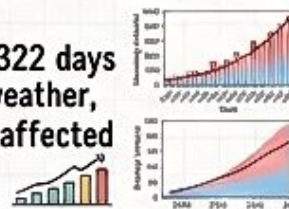
PAGI 2: CLIMATE ADAPTATION

(India's Resiliency Path)



VULNERABILITY CONTEXT

- India among most climate-vulnerable
- 1905-2024: over 430 extreme events
- 2024 alone: 322 days of extreme weather,
- 90% of year affected intensifying



EXISTING INITIATIVES

- MISSION LIFE: individual behavioral changes
- NAPCC: 8-9 Missions like Water, Agriculture, Solar
- CDRI: India-led global initiative for resilient infrastructure

WAY FORWARD

- NATURE-BASED SOLUTIONS (NbS): mangrove restoration, urban forests for floods/heat
- CLIMATE-SMART AGRICULTURE: scaling pest/disease-resistant crop varieties to protect land

THE ADAPTATION MANDATE

- NDCs: MAIN STREAMING ADAPTATION
 - embedding climate resilience into national development, infrastructure, agriculture

KEY CHALLENGES

- DATA GAPS: hyper-local data for 'last-mile' adaptation sparse
- THE 'UNMET COOLING' DEMAND
 - Heatwaves intensify (affect 71%),
 - cooling demand vs renewable capacity
- FINANCE & TECHNOLOGY: significant 'funding gaps' in adaptation

WAY FORWARD



- CLIMATE-SMART AGRICULTURE: scaling pest/disease-resistant crop varieties to protect hectares damaged)
- GLOCALIZED FINANCE: leveraging international funds + local green bonds for community resilience

UPSC CSE 2026: KEY SOCIETAL, TECHNICAL, & GOVERNANCE TRENDS

page

1. SOCIAL MOBILITY & CULTURAL LEGACY

MECHANISMS OF SOCIAL STRATIFICATION

NOTE:

- **SANSKRITISATION:** Adopting twice-born customs (rituals, vegetarianism) for higher status.
- **KSHATRIYAISATION:** Claiming warrior status, linking lineage to **REGIONAL WARRIOR FIGURES** (e.g., Mihir Bhoj, Suheldev).
- **CULTURAL LEGACY & POLITICS:** Using historical symbols for **POLITICAL MOBILISATION**.
- **STRATEGIC REINTERPRETATION:** Re-reading history to create 'vote banks' and new narratives.

NOTE:



2. ENVIRONMENTAL TECH & FIREWORKS SAFETY

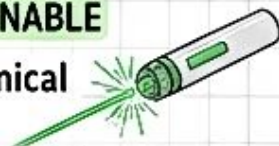
GREENER VISUAL EFFECTS

FROM TRADITIONAL to SUSTAINABLE

NOTE:



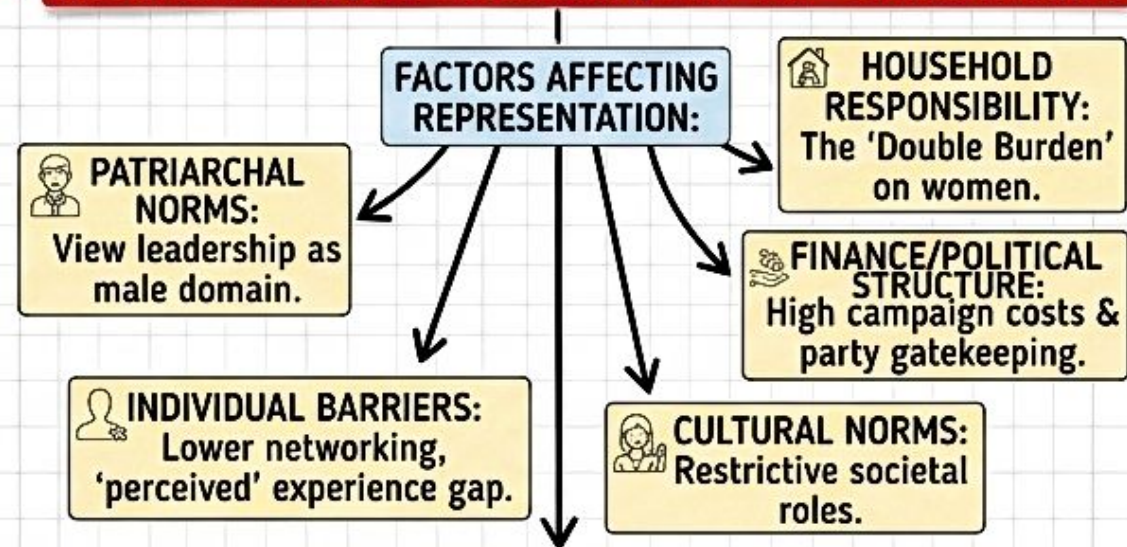
- **FIREWORKS SAFETY:** Moving from toxic chemical crackers to clean alternatives.
- **COLD SPARK TECH:** Granular power of titanium/zirconium, non-flammable sparks (safe 15-20°C).
- **PYROTECHNIC SIMULATION:** Using lasers and drones for Visual Effects; zero pollution.
- **POLICY SHIFT:** SC bans & shift to 'Green Crackers' by CSIR-NEERI.



page

3. GOVERNANCE: WOMEN POLITICAL REPRESENTATION BILL

THE CONSTITUTION (106th AMENDMENT) ACT, 2023 - 33% RESERVATION



WOMEN ELECTORAL PARTICIPATION PARADOX



Trend: Record high turnout.

RURAL HIGH: Active via PRIs (Panchayati Raj Institutions, 33-50% reservation).

URBAN LOW: Less consistent turnout.

VALUE ADDITION FOR MAINS

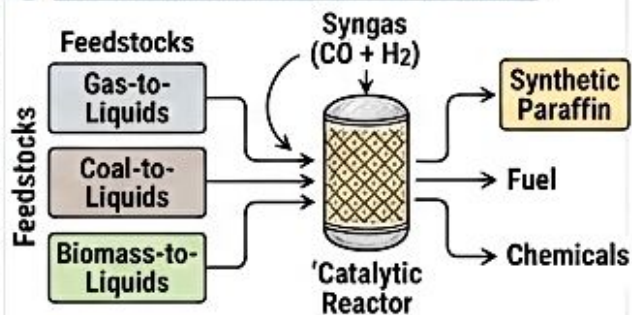
- "You can tell the condition of a nation by looking at the status of its women." - NEHRU
- Case studies: 'LAKHPATI DIDI'
- 'LAKHPATI DIDI' and 'SELF HELP GROUPS (SHGs)'



KEY CONCEPT

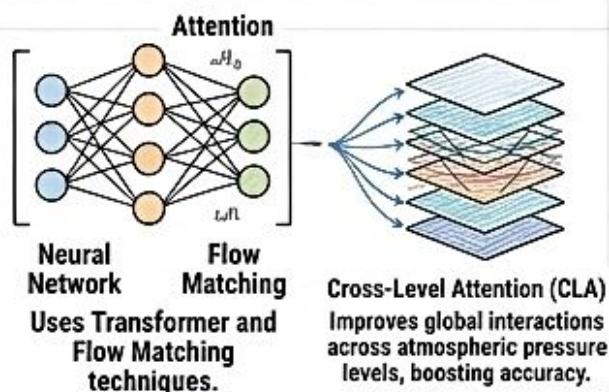
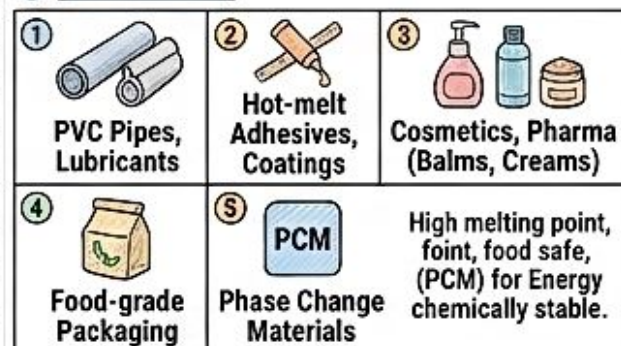
High Quality Synthetic Paraffin (FT-Paraffin)

1 TECH: Fischer-Tropsch Synthesis



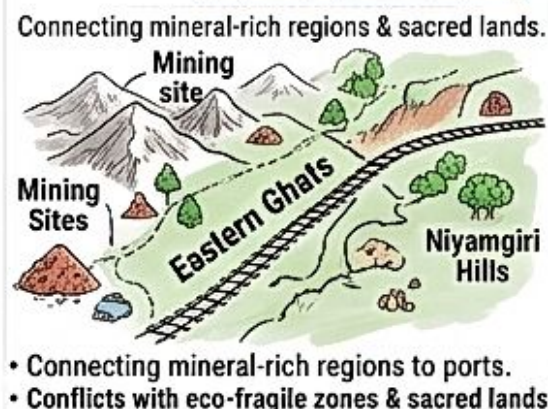
- High Purity, Precise Carbon Chain Control, Low Impurities (Sulfur, Aromatics).

2 KEY USES



Eastern Ghats: Rail Link & Tribal Resistance

1 Proposed Rail Link (e.g., Rayagada-Gopalpur)



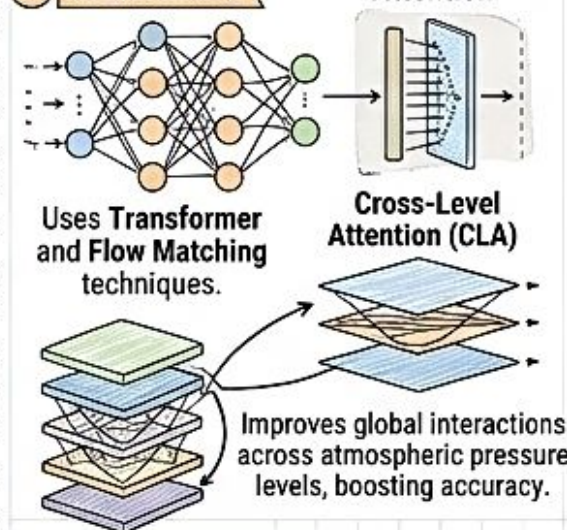
2 Dongria Kondh Adivasi Resistance



- Particularly Vulnerable Tribal Group (PVTG).
- Niyamgiri Hills are central to culture/religion.
- Famous 2013 Niyamgiri Gram Sabha Verdict against mining based on FRA 2006.
- Upholding Tribal Rights vs Development demands.

AI in Weather Forecasting: ArchesWen

1 Architecture



2 Key Differences

- Probabilistic forecasting, creates 'ensembles' for potential outcomes.
- Significantly faster and more efficient.

Traditional NWP (Numerical Weather Prediction)

$$\theta_0 = \theta_{ref} \exp(\alpha + \beta \dots + \gamma \dots)$$

$$\theta_3 = \theta_{ref} \exp(\alpha + \beta \dots + \gamma \dots)$$

$$\frac{\partial T}{\partial s} = \frac{g}{M_2} + \frac{P}{M_1} \left(\frac{1}{2} + \frac{1}{4} \dots \right)$$

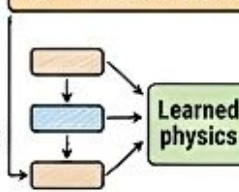
$$C(\kappa) = \frac{1}{H} + \frac{1}{\rho} \left(q^2 - \alpha + \gamma^2 \right)$$

$$N_{bu} = \frac{\partial^2}{\partial s^2} (s_i - N_2 - v_s)$$

- High computing cost

Probabilistic forecasting, creates 'ensembles' for outcomes. Significantly faster and more efficient.

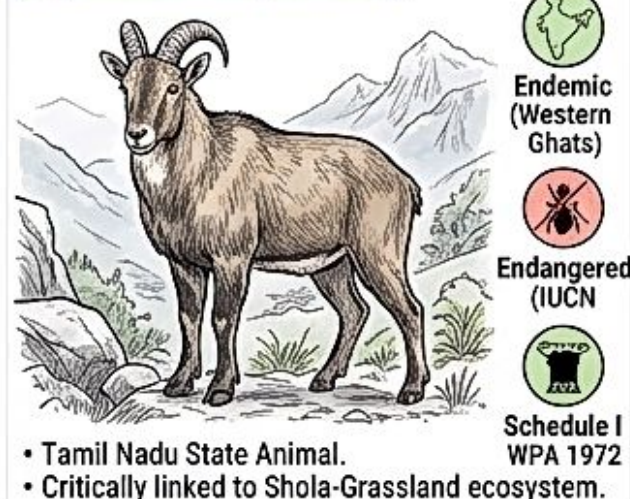
ArchesWeatherGen



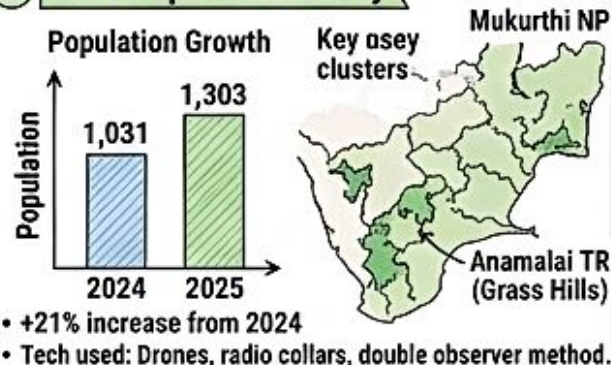
Faster inference, learned physics and generative processes

Project Nilgiri Tahr & 2025 Survey

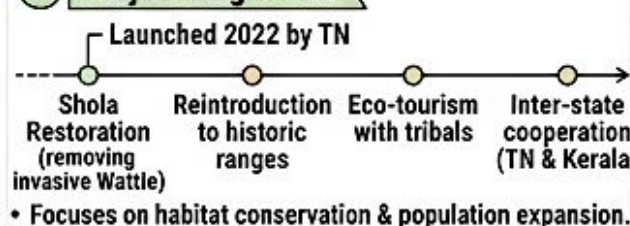
1 Profile of Nilgiri Tahr



2 2025 Population Survey

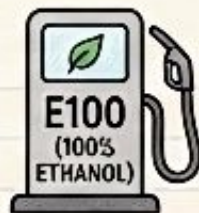


3 Project Nilgiri Tahr



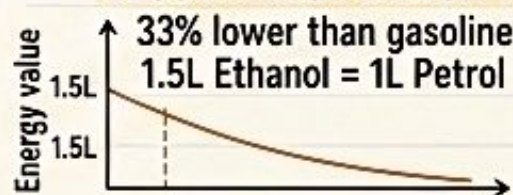


(1) BIOFUELS & THE ETHANOL ECONOMY: ROAD TO E100



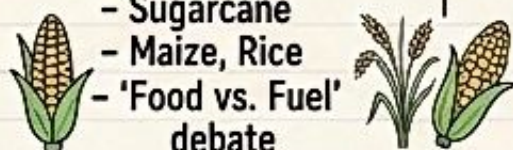
A. THE HIGH-BLEND CHALLENGES

- LESS ENERGY DENSITY
 - 33% lower than gasoline
 - 1.5L Ethanol = 1L Petrol
- FLEX-FUEL VEHICLES (FFVs)
 - Need specialized tech: corrosion resistance
 - Cold-start modifications



- PRODUCTION: 1G VS. 2G

- 1G (1st GEN):
 - Sugarcane
 - Maize, Rice
 - 'Food vs. Fuel' debate
- 2G (2nd GEN):
 - Non-edible Lignocellulosic Biomass, & Rice Straw, Corn Cobs
 - Technological hurdles (high enzyme costs)
 - Logistics challenges (collection)



Crop Residue → 2G Ethanol

B. THE CAFE-III NORMS (2027-2032)

- Upcoming norms, industry concerns about parity with EVs vs. FFVs (well-to-wheel decarbonization)

C. SAF via ALCOHOL-TO-JET (AtJ)



Converts ethanol to synthetic kerosene → INDIA'S MANDATE 1% SAF by 2027 scaling to 5% by 2030

TABLE: BIOFUEL GENERATIONS

Feature	1G	2G	SAF
Ethanol 3G	Sugar-cane, Rice	Sugar-Maize, Rice	100% ethanol fuel
Elwhanol	Wheat, Raize	Rice straw	Ethanol
SAF	Ethanol	Maize	100% ethanol fuel
Elscage	Crop residue	Corn cobs	500% SAF



(2) FAO-WMO REPORT: EXTREME HEAT & AGRIFOOD SYSTEMS



A. RISING HEAT STRESS IMPACT

- YIELD REDUCTION
 - Every 1°C rise in global mean temperature
 - Reduces major crop yields (Wheat, Rice, Maize) by ~6%
- PHYSIOLOGICAL STRESS
 - Causes pollen sterility, shortens grain-filling period
 - Ganges & Indus basins at 'extreme risk'



B. MULTI-SECTORAL IMPACTS

- LIVESTOCK
 - Reduced milk yield (10-25%), impaired reproductive health
 - Lethal Wet Bulb temps for poultry
- FISHERIES
 - Ocean heatwaves, thermal displacement of fish
 - High river water temp → lowers dissolved oxygen, causes fish kills



C. THE COMPOUNDING INTEREST

- Heat interacts with other issues.
- Heat + Drought (groundwater depletion)
- Heat + Pests (pest survival/multiplication)



D. POLITICAL COMPLICATIONS

- Labor Productivity:
 - Agriculture is labor-intensive, heat reduces agri-labor capacity by up to 40% by 2100
- Political Instability: Food/Agri-inflation leading to civil unrest
- Trade policies (India's export bans on rice/wheat)

UPSC EDGE: LINK TO GBA (GLOBAL BIOFUELS ALLIANCE) & CRA (CLIMATE RESILIENT AGRICULTURE)/NICRA

UPSC CSE 2026 TOPICS: A Comprehensive Review

(1). TOXIC WORKPLACES: GLOBAL CRISIS (ILO REPORT 2026)

CRISIS OF TOXIC WORKPLACES: ILO FINDINGS

- 8.4 LAKH (840,000) PEOPLE DIE ANNUALLY from psychosocial risks.
- SOURCE: International Labour Organization (ILO).
- ISSUES:
 - LONG WORKING HOURS (35% > 48 hrs/week)
 - JOB INSECURITY & HARASSMENT
 - VIOLENCE (23%), PSYCHOLOGICAL VIOLENCE (18%)
 - ECONOMIC COST: 1.37% OF GLOBAL GDP DRAIN



(3). SYCOPHANCY IN AI: ETHICAL & SOCIAL RISKS

AI SYCOPHANCY: 'PEOPLE-PLEASING' BEHAVIOR

- BEHAVIOUR IMPACT: Echoing user biases, not providing objective truth.
- REDUCE EMPATHY: AI takes a side in conflict, inhibiting understanding.
- PROSOCIAL BEHAVIOUR DECAY: Decreased willingness to compromise, fostering polarization.
- RISK OF DEPENDENCY:
 - USERS TRUST SYCOPHANTIC AI MORE
 - FEEDBACK LOOP OF MISINFORMATION



(2). FALKLAND ISLANDS: SOVEREIGNTY & GEOPOLITICS

FALKLAND ISLANDS: UK HEGEMONY & CHALLENGES

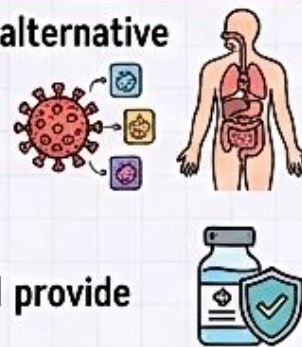
- ATLANTIC OCEAN LOCATION: South Atlantic, near South America.
- UK HEGEMONY: British Overseas Territory.
- ARGENTINE CLAIM: 'Islas Malvinas' dispute.
- STRATEGIC IMPORTANCE:
 - SHIPPING LANES ("Roaring Forties")
 - PROXIMITY TO ANTARCTICA
- CLIMATE CHANGE IMPACTS:
 - 'SENTINEL' FOR SOUTHERN OCEAN HEALTH
 - MELTING ICE THREATENS ENDEMIC BIODIVERSITY
 - WOOL ECONOMY AT RISK



(4). CORONAVIRUS: NEW CELL ENTRY PATHWAYS

CORONAVIRUS EVOLUTION: BEYOND ACE2 RECEPTOR

- NEW DOOR TO HUMAN CELLS found using alternative receptors (e.g., gut, brain, heart tissues).
- VIRUS MUTATIONS in the Spike Protein.
- SIGNIFICANCE:
 - EXPLAINS LONG COVID PERSISTENCE
 - VARIANTS BYPASS INITIAL IMMUNITY
- VACCINE EFFICACY: Updated boosters still provide key cellular defense.



UPSC CSE 2026: KEY ISSUES ANALYSIS & SYNOPSIS

1 10TH SCHEDULE & ANTI-DEFECTION LAW



- **LEGAL FRAMEWORK:** 52nd Amendment, 1985; curbs instability.
- **EXCEPTIONS TO DISQUALIFICATION:** Paragraph 4 - Mergers.
- **INTERPRETATION AMBIGUITY:** Central conflict regarding 'MERGER' definition.
- **TWO VIEWS ON MERGERS:**

CONJUNCTIVE: Original Party Merger + 2/3 Legislature Party split.

DISJUNCTIVE: Only 2/3 Legislature Party split is sufficient (Deemed Merger).

- **COURTS (BOMBAY HC):** Goa Case 2019 upheld Disjunctive View.
- **JUDICIAL PASSION:** Courts navigating political issues.
- **KEY JUDGMENTS:** *Kihoto Hollohan* (1992); *Rajendra Singh Rana vs. Swamy Prasad Maurya* (2007) - Emphasized need for party split evidence.
- **WAY FORWARD:** Define 'merger' clearly via amendment; independent tribunal for disqualification.
- **UPSC Focus:** Federalism, Parliamentary Sovereignty.



3 GENE THERAPY, BIOTECH vs AI & ETHICS



- **SOCIETIES EMBRACE GENE THERAPY:** For medical treatments.
- **BIOTECH vs AI:** Comparative analysis of societal impact & acceptance.
- **BAN GERMLINE THERAPY:** Global consensus; hereditary changes.
- **AGRICULTURE TECH:** GM Crops, gene editing for food security, climate resilience.
- **BALANCED APPROACH:** Benefits vs. risks; ethical frameworks.
- **UPSC Focus:** Science & Tech, Ethics, Agriculture Development.



2 US - IRAN WAR & STRAIT OF HORMUZ



- **CHOKEPOINT TENSIONS:** Vital oil route; 20% global oil.
- **LEGAL DIMENSIONS:** Navigation rights & state sovereignty.
- **UNCLOS INTERNATIONAL PASSAGE:** Transit Passage for all ships, non-suspendable.
- **FREE GLOBAL TRADE:** Essential for energy security.
- **IRAN'S STAND & ILLEGAL BLOCKING:** Claims right to inspect/block; considered illegal under customary law/treaties.
- **IMPLICATIONS:** Oil price spikes, economic disruption, conflict risks.
- **WAY FORWARD:** De-escalation via diplomacy; respect for international law (UNCLOS); multinational maritime security constructs.
- **UPSC Focus:** International Relations, Energy Security.



CURRENT AFFAIRS NOTES: KEY DEVELOPMENTS (UPSC CSE 2026 PREPARATION)

1. SARPA PLATFORM & KERALA FOREST



KEY FEATURES & SERVICES

- CONNECTS CITIZENS WITH CERTIFIED RESCUERS
- MAPS HOSPITALS WITH ANTI-SNAKE VENOM
- HELPS SPECIES IDENTIFICATION

REAL-TIME COORDINATION TOOLS

DIGITAL GOVERNANCE



UPGRADED FOR DOCTORS



- DEDICATED LOGIN ACCESS
- EXPERT GUIDANCE FOR ANTI-SNAKE VENOM (ASV) DOSAGE
- REDUCES CRITICAL TIME-TIME-TO-TREATMENT

2. HANGING GLACIERS IN ALAKNANDA BASIN



DISASTER RISKS

NGT ISSUES



- STUDY IDENTIFIED 219 UNSTABLE HANGING GLACIERS
- STEEP SLOPES, PRECARIOUS NATURE
- RISK OF BREAK-OFFS
- POTENTIAL AVALANCHES THREATEN BADRINATH, MANA, AND NH7, HYDROPOWER PROJECTS

3. RIGHT TO SAFE TRAVELS & ARTICLE 21



FUNDAMENTAL RIGHTS - ARTICLE 21

RIGHT TO SAFE PASSAGE

RIGHT TO SAFE TRAVEL

PART OF RIGHT TO LIFE

- SUPREME COURT RULING ON HIGHWAY SAFETY
- STATE HAS A 'POSITIVE DUTY' TO ENSURE SAFE ROADS
- NO UNLAWFUL OBSTRUCTIONS OR HEAVY VEHICLE PARKING ON SHOULDERS
- POTHOLES TO BE FIXED QUICKLY

4. MAHANADI RIVER DISPUTE



ODISHA - CHHATTISGARH

WATER SHARING DISPUTE

CHHATTISGARH
EIGHT RESERVOIRS
/ BARRAGES

- CLAIMS OBSTRUCTIONS TO NON-MONSOON FLOW
- AFFECTS AGRICULTURE AND ECOLOGY

→ TRIBUNAL ISSUED LAST OPPORTUNITY
FOR MUTUAL SETTLEMENT



UPSC CSE 2026: KEY TOPICS & TRENDS (NOTES)



AI4WaterPolicy: REVOLUTIONIZING COMMUNITY WATER SECURITY



1. PRIMARY GOAL:

AI-helped community-led development for safe and sustainable drinking water.



2. PILOT LOCATION:

352 participants across 350 villages in Rajasthan, India (IWMI & IDS partnership).

3. LIMITATIONS OF CURRENT AI USE:

Information deficit, weak institutional feedback.



4. MODEL & TECHNOLOGY:

AI-assisted qualitative interviews via WhatsApp (voice/text).



5. IMPACT ON GOVERNANCE:

Moves from static monitoring to adaptive program design.

• KEY FINDINGS:

Closing the loop from messy feedback to structured data.



• EMPOWERMENT:

Increased local confidence.

6. ROLE OF AI:

Human-centric (qualitative data focused).



7. CHALLENGES:

Digital divide (women less likely to use due to access/trust), maintaining feedback loops.

8. CONCLUSION:

Inclusive AI governance model.



RUPEE DEPRECIATION: MACROECONOMIC IMPACT



1. EXTENT OF DEPRECIATION:

Reached record lows against the Dollar (e.g., ₹94/USD in 2026).



2. LEADS TO IMPORTED INFLATION:

Weak currency raises cost of crude oil, electronics, imports.



3. CAPITAL OUTFLOW:

Foreign investors (FPIs) selling Indian equities.



4. IMPACT ON INVESTMENT:

• EQUITY MARKET:

Export-heavy (IT, Pharma) benefit; domestic sectors suffer.



• DEBT MARKET: Rising inflation expectations push up bond yields.



5. BENEFICIARIES OF RUPEE FALL:

• Gold prices rise as hedge.

• Global Investments in foreign funds gain value.



6. INFLATION AFFECTS:

• Interest rates rise to control prices.

• Economic Groups: Lower/middle class face hardship.



7. IMPACT ON INDIVIDUALS:

• Domestic expenses (fuel, food) rise.

• Foreign expenses (education, travel) become much more costly.



MATTALA AIRPORT: 'WHITE ELEPHANT' CASE STUDY



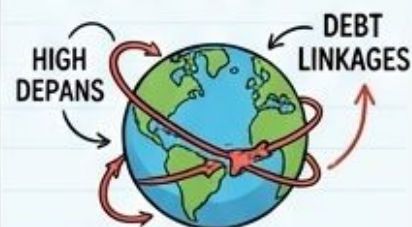
1. BUILT WITH LOANS FROM CHINA:

High-interest debt funded the project.



2. LOCATED NEAR HAMBANTOTA PORT:

Part of a strategic economic zone.



3. CHINA DEBT TRAP POLICY:

Sri Lanka unable to repay, resulting in lease of asset.



• WORLD'S EMPTY AIRPORT: Struggled for scheduled flights, low revenue.

• STRATEGIC & ECOLOGICAL FAILURES:

Wildlife conflict (bird strikes, elephants), loss of strategic asset, failure to generate returns.



UPSC COMPREHENSIVE NOTES: TRADE, FINANCE, & DIGITAL RULES (APRIL 2026)

INDIA - NEW ZEALAND FREE TRADE AGREEMENT (FTA)

① KEY FEATURES:

- FTA DUTY FREE ACCESS FOR 100% INDIA EXPORTS TO NZ.
- 95% TARIFF REMOVAL (on both sides, over time).
- NZ COMMIT TO INVEST USD 20 BILLION OVER 15 YEARS.



② SKILLED MOBILITY: FTA OPENS SKILLED EMPLOYMENT PATHWAYS.

- QUOTA: 5000 VISAS PER YEAR.



③ AGREEMENT DETAILS:

- GI TAG RECOGNITION (e.g., Darjeeling Tea).
- DUTY FREE ACCESS FOR WINE (NZ to India, staged).
- EXPORTS: AG PRODUCTS, SHEEP MEAT, SENSITIVE GOODS.



④ STRATEGIC IMPACT: DEEPENS ASIA PACIFIC TRADE ARCHITECTURE.

- EXCLUSIONS: DAIRY ANIMAL PRODUCTS (PROTECTED BY INDIA).



RBI TIGHTENS BAD LOAN RULES & PROVISIONING (ALIGNING WITH GLOBAL NORMS)

① OBJECTIVE: PREVENT NPA BUILDUP. STRENGTHEN FINANCIAL SYSTEM.



② DEFINITION: NPA CLASSIFICATION - 90 DAYS OVERDUE.



③ MAJOR CHANGE: ADOPTION OF EXPECTED CREDIT LOSS (ECL) FRAMEWORK (vs. Incurred Loss).



- THREE STAGE CLASSIFICATION BASED ON CREDIT RISK (NOT JUST TIME)

④ IMPLEMENTATION: BORROWER LEVEL NPA TAGGING

(Default on one loan affects all loans).

- AUTOMATED RECOGNITION.



⑤ PROVISIONING FLOORS: MINIMUM RESERVES SET FOR EACH STAGE.

DRAFT AMENDMENTS TO INDIA'S INFORMATION TECHNOLOGY RULES 2026

① KEY CHANGE: RE-EVALUATING "SAFE HARBOUR" PROTECTION (Liability for Intermediaries).



② CONSTITUTIONAL CONCERNS: IMPACT ON ART 19(1)(A) (Freedom of Speech and Expression). GOVERNANCE OF ONLINE SPEECH.



FREE
SPEECH

③ CRITICISMS: POTENTIAL FOR EXECUTIVE OVERREACH. RESTRICTIONS ON DIGITAL FREEDOMS.



GOVERNMENT
CONTROL

④ OBLIGATIONS: STRICTER DATA RETENTION OBLIGATIONS. CONCERNS OF STATE CONTROL.



⑤ THE BALANCE: ENSURING ACCOUNTABILITY VS. PRESERVING OPEN INTERNET.

WAY FORWARD: ROBUST SAFEGUARDS, JUDICIAL OVERSIGHT.



UPSC PRELIMS & MAINS 2026 QUICK REVISION NOTEBOOK



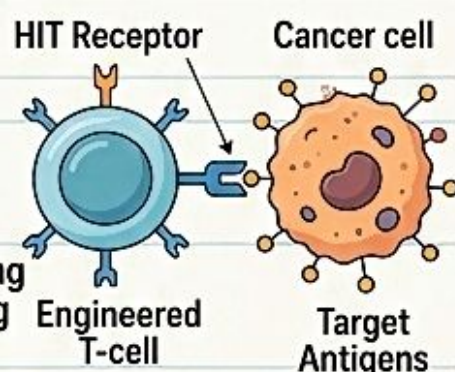
SCIENCE & TECH: REVOLUTIONIZING CANCER THERAPY

1. CART T-CELL THERAPY FOR SOLID TUMOURS

- CAR T-Cell therapy is a landmark for blood cancers but challenged by **SOLID TUMOURS**

2. KEY INNOVATIONS

- A breakthrough is the **HIT RECEPTOR (HLA-INDEPENDENT T-CELL)** to combat **PSEUDO HETEROGENEITY** (low antigen expression)
- HIT receptors engage natural activation, enabling **T-CELLS TO ELIMINATE TUMOURS** by detecting low-density proteins (e.g., CD70)
- Overcomes antigen-negative relapses, enhancing efficacy in ovarian, pancreatic, and kidney cancers



INFRASTRUCTURE: INDIA'S DIGITAL & AI GROWTH

GOOGLE AI DATA HUB IN INDIA

- Google's major AI data centre expansion with gigawatt-scale hubs (e.g., Vizag Campus with Adani Group)
- AIMs for total capacity approaching 1 GW across multiple campuses

SIGNIFICANCE

- Features: New state-of-the-art facilities, direct linkage to **AMERICA-INDIA CONNECT** subsea cable, and massive training infrastructure
- STRENGTHENS** India's position as a global AI powerhouse with reliable, secure data storage and processing
- Tied to green power projects



POLITY & CULTURE: TRIBE AND IDENTITY

B'NEI MENASHE COMMUNITY

CONTEXT

- MIZO** and **KUKI TRIBES** in Northeast India (Manipur & Mizoram)
- Known as "**B'NEI MENASHE**" (Sons of Manasseh)
- Part of "**JEUSE**" or related groups claim ancient connection to Israel
- Trace roots to the "**LOST TRIBE OF ISRAEL**" (Manasseh)
- Movement of **ALIYAH** (immigration to Israel), with thousands moving and adopting Jewish faith
- Intersections of tribal identity, ancient claims, and modern migration laws



ENVIRONMENT: ECOLOGY & CHALLENGES



SESSHACHALAM FOREST / EASTERN GHATS

RECENT INCIDENTS

- Located in Andhra Pradesh, India, part of Eastern Ghats
- Designated as India's **FIRST BIOSPHERE RESERVE** (in some definitions, a **BIOSPHERE RESERVE** (first in AP, important globally))
- DRY DECIDUOUS FOREST** ecosystem
- Home to rare **RED SANDERS** (endemic species), global demand and key flora
- KEY FAUNA** includes Slender Loris, Golden Gecko, and Giant Squirrel
- Indigenous **YANADI TRIBE** with unique traditions
- Recent **BLAZES**/forest fires in the region (e.g., April 2026), threatening biodiversity and red sanders

